

An Introduction to the Representation Theory of Lie Groups and the Orbit Method

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Chapter 1

Introduction

"The chief forms of beauty are order
and symmetry"

Aristotle, Metaphysics book XIII

A central object of harmonic analysis is the space of square integrable functions $L^2(G)$ where G is some group. The theory of classical Fourier series is recovered when $G = \mathbb{T}$. Of key interest is finding a basis for $L^2(G)$; in the case of $L^2(\mathbb{T})$ the answer is given by the set of functions $\{e^{2\pi i n x} \mid n \in \mathbb{Z}\}$, in the language of representation theory we refer to the functions as the *characters*. The Fourier transform of a function in $L^2(\mathbb{T})$ is given by

$$\hat{f}(n) = \int_0^1 f(x) e^{-2\pi i n x} dx,$$

we may then write the inverse Fourier transform

$$f(x) = \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{2\pi i n x},$$

where the convergence is in L^2 . Notice the domain of $\hat{f}$ is $\mathbb{Z}$ and the domain of f is $\mathbb{T}$, implying a notion of duality $\mathbb{Z} \longleftrightarrow \mathbb{T}$ called *Pontryagin duality*. Plancherel proved that the Fourier transform is a unitary isometric transformation, and so constitutes a Hilbert space isomorphism

$$L^2(\mathbb{T}) \cong \ell^2(\mathbb{Z}),$$

and for $f \in L^2(\mathbb{T})$

$$\|f\|_{L^2} = \|\hat{f}\|_{\ell^2}.$$

The history of harmonic analysis traces the extension of these concepts to other groups; a key first step is the extension to $G = \mathbb{R}$. This already poses significant analytical challenges

that prefigure the development of harmonic analysis in the movement from a compact to a non-compact domain. For a function $f \in L^2(\mathbb{R})$, the Fourier transform is defined

$$\hat{f}(\xi) = \int f(x) e^{-2\pi i \xi x} dx \quad \xi \in \mathbb{R},$$

and the inversion formula

$$f(x) = \int \hat{f}(\xi) e^{2\pi i \xi x} d\xi,$$

where the sum in the compact case becomes an integral, indicating the shift in domain of $\hat{f}$ from $\mathbb{Z}$ to $\mathbb{R}$. This is an example of Pontryagin duality for locally compact abelian groups. The study of Fourier theory in L^2 is devoted to the convergence conditions of these integrals where we recover the Plancherel formula

$$\|f\|_{L^2} = \|\hat{f}\|_{L^2}.$$

Given the above we say that the Fourier transform 'preserves the L^2 norm of f ', allowing us to pass problems about f to problems about $\hat{f}$ in cases that concern only the L^2 properties of f . This technique has proved extraordinarily useful in mathematics, engineering and the sciences more broadly.

Many physical problems cause us to consider functions on manifolds other than $\mathbb{R}$, functions on the sphere S^2 for example. To understand these functions we would like to extend the powerful tools of Fourier theory to these domains. In order to do so we must reformulate Fourier theory in the language of representation theory. A representation of a group G is a homomorphism

$$\rho : G \rightarrow \text{GL}(V).$$

This representation is *irreducible* if there does not exist any non-zero proper subspace of V invariant under the action of G defined by the representation. The representation is *unitary* if $\rho(g)$ defines a unitary operator for all $g \in G$. Write $\hat{G}$ to denote the set of all unitary irreducible representations of G . For a compact group G we develop a suitable measure dg called the *Haar measure* which allows us to form $L^p(G)$ spaces where the p norm is defined with respect to dg . As in the case of $G = \mathbb{T}$ we would like to find a basis for this space, here the Peter-Weyl theorem delivers the answer

Theorem 1.0.1. *Let G be a compact group, with unitary dual $\hat{G}$, then*

$$L^2(G) = \bigoplus_{\pi \in \hat{G}} d_\pi V^\pi$$

where V^π denotes representation associated with the representation π , and d_π the multiplicity.

For $\mathbb{T}$ (and in fact any abelian group) the characters form the basis for this space and the Fourier transformation has the natural definition

$$\hat{f}(\chi) = \int f(g) \overline{\chi(g)} dg.$$

In the case of a locally compact abelian group, the unitary dual $\hat{G}$ is again a topological group, Plancherel's theorem applied to LCA groups then provides a measure η on $\hat{G}$ such that

$$\int_G |f(x)|^2 dg(x) = \int_{\hat{G}} |\hat{f}(\xi)|^2 d\eta(\xi)$$

holds, and so

$$f(x) = \int_{\hat{G}} \hat{f}(\xi) e^{2\pi i \xi x} d\eta(\xi)$$

is defined for a dense subset of $L^2(G)$. In the case of a general compact group, the *matrix coefficients*

$$M_\pi = \{f_{v,w}(x) = \langle \pi(x)v, w \rangle \mid v, w \in V\}$$

(where π is a representation of G on a vector space V) assume the role of characters.

Of particular physical and analytical interest are Lie groups which can be compact or non-compact. The problem of forming a representation theory for non-compact semisimple Lie groups was the focus of Harish-Chandra who developed a Plancherel formula for connected reductive groups. This is a complex and deep theory constituting a life of work in mathematics and will not be presented here.

We will however discuss the development of harmonic analysis for non-compact nilpotent groups. At this juncture we are led directly to Kirillov theory which establishes the representation theory for nilpotent Lie groups through the development of the orbit method. Kirillov uses results from Dixmier and Mackey in his proof describing the unitary dual $\hat{G}$ and provides an analog of the Plancherel formula, establishing the representation theory and harmonic analysis for nilpotent Lie groups.

The orbit method is a philosophy which states that (broadly) the study of the representation theory of a group reduces to the study of the coadjoint orbits. An introduction of the orbit method was realised by Kirillov in [7] centred on the study of nilpotent Lie groups. The orbit method was implicitly present in the work of Mackey [4] in the restriction of the 'Mackey machine' to Lie groups G that can be written as a semidirect product of a closed Lie subgroup H and an abelian group N such that $G = H \rtimes N$.¹ Broadly the orbit method looks something like

- Classify the coadjoint orbits $\Omega \in \mathcal{O}(G)$ in the dual of the Lie algebra $\mathfrak{g}^*$
- Any $\beta \in \Omega$ gives rise to a character ρ of the stabiliser subgroup G_β :

$$\rho(\exp X) = e^{2\pi i \langle \beta, X \rangle}$$

- The irreducible representation of the whole group is then given by the induced representation:

$$(\text{Ind}_{G_\beta}^G \rho)(g). \tag{1.1}$$

¹ See that for an abelian group N , that $N \cong N^* \cong \hat{N}$, so coadjoint orbits are equivalent to orbits in the unitary dual.

There are a number of caveats to this simplified presentation depending on the structure of the group being considered, this thesis will outline these caveats and provide the necessary background in order to understand this construction.

Kirillov's original presentation of the orbit method in [7] relies on the algebraic properties of nilpotent Lie algebras, along with the induced representation machinery of Mackey. It turns out however that the geometric properties of the orbits considered as submanifolds of $\mathfrak{g}^*$ are of great interest in service of the generalisation of the orbit method to a broader class of groups. The key geometric property one exploits is that every coadjoint orbit is a *symplectic manifold*. Symplectic manifolds M are the natural setting for Hamiltonian mechanics where one defines a Hamiltonian

$$H : M \rightarrow \mathbb{R},$$

which induces a vector field X_H on M , the integral curves of which will define level curves of H . Physically these level curves are a collection of states where energy is conserved. Geometric quantisation aims to develop a representation of the Poisson algebra of smooth functions $f \in C^\infty(M)$ on a Hilbert space, namely the space of sections $\Gamma(L)$ of a line bundle L over M . A coadjoint orbit has a natural symplectic structure given by the Kirillov-Kostant-Souriau two-form, this is the key object which translates between the algebraic properties and the geometric properties of the orbit. The group G acts on a coadjoint orbit $\Omega \in O(G)$ via the coadjoint action. As above, the theory of geometric quantisation finds a representation

$$C^\infty(\Omega) \rightarrow \Gamma(L).$$

Given such a representation, some $X \in \mathfrak{g}$ and $m \in \Omega$, we can write the function

$$X(m) := \langle m, X \rangle,$$

(where $\langle m, X \rangle$ is the standard pairing of an element with the dual). These functions form a Lie subalgebra of $C^\infty(M)$ which is isomorphic to $\mathfrak{g}$. In this way we find a representation of the Lie algebra.

After introducing a polarisation to reduce the size of the space $\Gamma(L)$, we are left with the space $\Gamma_{hol}(L)$ of holomorphic sections, where the action of G is unitary and irreducible. This space is in agreement with the theorem of Borel-Weil where the coadjoint orbits appear as flag manifolds, in this way the orbit method is a generalisation of Borel-Weil.

The orbit method is not one piece of mathematics, it is an ideology that claims a correspondence between coadjoint orbits and unitary irreducible representations, from this conviction one may develop and recover a representation theory for different kinds of groups. This thesis aims to add detail into the picture presented above through computing the examples of the Heisenberg group and $SU(2)$. Some proofs have been offered when appropriate however in efforts to convey the ideas and impetus of the theory the bulk of this work is devoted to definitions, intuitions and computations.

Chapter 2

Lie Algebras and Lie Groups

"Reality is partial to symmetry and
slight anachronisms"

Jorge Luis Borges

This chapter provides elementary definitions and explanations regarding Lie groups. Broadly, Lie groups encode continuous symmetries affording the quotienting of complex problems into simpler parts. We begin with a chain of definitions that provide context for the ubiquity of these objects.

Definition 2.0.1 (Topological Manifold,[33, pp. 3]). A topological space M is called an n -dimensional topological manifold if it satisfies the following three properties:

1. Hausdorff: For every pair of distinct points $p, q \in M$, there exist disjoint open subsets $U, V \subseteq M$ such that $p \in U$ and $q \in V$.
2. Second-countable: The topology of M has a countable basis.
3. Locally Euclidean: For every point $p \in M$, there exists an open neighborhood U of p that is homeomorphic to an open subset of $\mathbb{R}^n$.

A homeomorphism from such a neighbourhood U of $m \in M$ to $\mathbb{R}^n$ is called a *chart*.

Definition 2.0.2. Given two charts (U, ϕ) and (V, ψ) , where $U \cap V \neq \emptyset$. The *transition map* $\psi \circ \phi^{-1}$ is defined on the intersection.

Definition 2.0.3. A smooth manifold M is a manifold where the transition maps are C^∞ .

The structure of a smooth manifold makes possible the application of calculus on the manifold. For any smooth manifold M we can develop a useful notion called the *tangent space*. This is the set of equivalence classes of tangent vectors to curves through a point

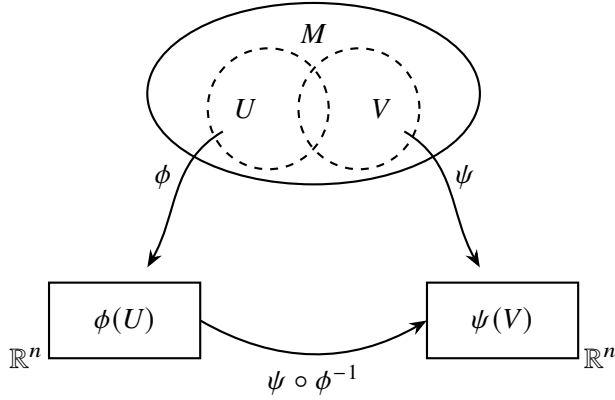


Figure 2.1: Transition maps

$m \in M$. The tangent space at a point m is written as $T_m M$, and forms a local approximation to the manifold at the point. The tangent bundle TM :

$$TM = \bigsqcup_{m \in M} T_m M,$$

is a fibre bundle over the manifold M . This means there is a projection

$$\begin{aligned} \pi : TM &\rightarrow M \\ T_m M &\mapsto m \end{aligned}$$

This setting makes it possible to characterise vector fields (functions that assign a vector to every point in M) on M as sections of the fibre bundle TM . Sections are maps

$$s : M \rightarrow TM,$$

such that $p \circ s$ is the identity map, and the element $\pi^{-1}(m)$ is identified with the tangent space $T_m M$ called the fibres of the bundle. The elements of the tangent space $T_m M$ are equivalence classes of tangent vectors to curves through a point m .

Definition 2.0.4. A topological group G is a topological space with a group operation $G \times G \rightarrow G$ such that $(x, y) \mapsto xy^{-1}$ is continuous with respect to the topology.

Definition 2.0.5. A Lie group G is a smooth manifold that is also a topological group where the group operation $G \times G \rightarrow G$, $(x, y) \mapsto xy^{-1}$ is C^∞ .

Example 2.0.1. Here are some commonly encountered Lie groups

- $\mathbb{T}^n$
- $\mathbb{R}^n$

- $U(n)$ the group of $n \times n$ unitary matrices
- $GL_n(\mathbb{C})$ the group of linear transformations of dimension n

In fact we have

Theorem 2.0.1 (Cartan's theorem). *Any closed subgroup H of a Lie group G is again a Lie group.*

Corollary 2.0.1.1. *Any closed subgroup of $GL_n(\mathbb{C})$ is a Lie group.*

We thus define

Definition 2.0.6. A matrix group over a field $\mathbb{C}$ is a closed subgroup of $GL_n\mathbb{C}$

The converse of corollary 2.0.1.1 is not true, consider the metaplectic group $Mp_{2n}(\mathbb{R})$ which has no faithful finite dimensional representations, so cannot be written as a matrix group. The remainder of this thesis will focus predominantly on matrix Lie groups.

2.1 Topological properties

Connectedness

We further characterise Lie groups by their topological properties, namely connectedness.

Definition 2.1.1. A Lie group G is connected if it is connected as a topological space, that is it cannot be written as the union of two or more non-empty disjoint open subsets.

Definition 2.1.2. A Lie group G is path connected if for any two points $x, y \in G$ there is a continuous function $f : [0, 1] \rightarrow G$ parameterising a line between the two points.

Definition 2.1.3. A path connected Lie group G is simply connected if the fundamental group $\pi_1(G)$ is trivial.

The fundamental group $\pi_1(G)$ is the set of all loops starting and ending at a point $y \in G$ considered equivalent up to homotopy. Given two loops γ_1, γ_2 starting and ending at a point $y \in G$, if γ_1 can be continuously deformed into γ_2 , then the two loops are considered homotopy equivalent.

Compactness

Definition 2.1.4. A topological group is compact if every open cover admits a finite subcover. For all open coverings $\{U_i\}_{i \in I}$ there exists a finite subset $J \subset I$ such that

$$M = \bigcup_{i \in J} U_i$$

2.2 Lie algebras

The Lie algebra corresponding to a Lie group is a linear object we can use to study Lie groups, which are often non-linear and complicated. Morally, the Lie algebra encodes the local information about a Lie group. We begin with some definitions.

Definition 2.2.1. A Lie algebra A is a vector space endowed with a Lie bracket $[\cdot, \cdot] : A \times A \rightarrow A$.

The bracket $[\cdot, \cdot]$ must be bilinear, alternating and satisfy the Jacobi identity. All Lie algebras are abelian with respect to the addition in the vector space, however we call a Lie algebra abelian when $[A, A] = 0$. Similarly a Lie algebra is nilpotent when it has a vanishing lower central series

$$A \supset [A, A] \supset [[A, A], A] \supset \cdots = 0,$$

and solvable when

$$A \supset [A, A] \supset [[A, A], [A, A]] \supset \cdots = 0.$$

A Lie algebra is *simple* if it is non-abelian and contains no ideals (not including 0 and the group itself) and *semisimple* if it can be written as a direct sum of simple Lie algebras.

2.3 Lie algebra of a Lie group

Given a Lie group G we can construct a Lie algebra $\mathfrak{g}$ given by the set of all left translation invariant vector fields on G . Let

$$\begin{aligned} L_g : G &\rightarrow G \\ x &\mapsto gx \end{aligned}$$

and denote the action of translation by g . The differential of L_g is denoted $(L_g)_*$ is a map on the tangent bundle

$$(L_g)_* : TG \rightarrow TG.$$

Evaluated at a point h , this forms a map between tangent spaces

$$(L_g)_{*,h} : T_h G \rightarrow T_{gh} G.$$

i.e. The tangent space at h is mapped to the tangent space at gh . A vector field F on G is left translation invariant if

$$(L_g)_{*,h} F_h = F_{gh}$$

In alternate notation: Given an element $X \in \mathfrak{g}$, the unique left invariant vector field associated to X is given by

$$\hat{X}(g) = \frac{d}{dt} L_g \exp(tX) \Big|_{t=0} = \frac{d}{dt} g \exp(tX) \Big|_{t=0} = g \cdot X.$$

The set of all such vector fields is closed under the Lie bracket of vector fields and hence forms a Lie algebra. This set can be associated with the tangent space at the identity $T_e G$. Any vector field can be understood as a differential operator, given $f \in C^\infty(G)$,

$$(\hat{X}f)(g) = \frac{d}{dt}f(g(\exp(tX)))|_{t=0}.$$

Choosing some local coordinates $(x_1, \dots, x_n)$ on the Lie group, the tangent space at a point will be spanned by the terms $\frac{\partial}{\partial x_i}$, we write the vector field as

$$\hat{X} = \sum_{i=1}^n \alpha_i \frac{\partial}{\partial x_i},$$

and write the equivalence of one forms

$$(\hat{X}f)(g) = df_g(\hat{X}(g)).$$

2.4 Lie Group - Lie Algebra correspondence

Define a function that maps the Lie algebra to a Lie group, we call this map the exponential map and write:

$$\begin{aligned} \exp : \mathfrak{g} &\rightarrow G \\ X &\mapsto \alpha(1), \end{aligned}$$

where α is the unique one parameter subgroup of G such that $\frac{d}{dt}\alpha(t)|_{t=0} = X$.

Definition 2.4.1. A one parameter subgroup of a Lie group G is a function defining a curve in G given by

$$\alpha : \mathbb{R} \rightarrow G,$$

such that $\alpha(s+t) = \alpha(t)\alpha(s)$, i.e. α is a homomorphism.

Define the one parameter subgroup α as the solution to

$$\alpha'(t) = \hat{X}(\alpha(t)),^1$$

where $\alpha(0) = e$ and $\hat{X}$ is the unique left invariant vector field corresponding to $X \in \mathfrak{g}$. Notice that this says that the vector tangent to the curve $\alpha(t)$ is always $\hat{X}(\alpha(t))$. Furthermore,

$$\alpha'(0) = X.$$

¹This curve is often referred to as the integral curve of $\hat{X}$.

The last property we verify is that α is a homomorphism

$$\begin{aligned}\phi(s) &= \alpha(s+t) \\ \phi_2(s) &= \alpha(s)\alpha(t).\end{aligned}$$

Both these curves are integral curves for $\hat{X}$ and hence equivalent,

$$\phi(s) = \phi_2(s) = \alpha(s+t) = \alpha(s)\alpha(t).$$

When G is a matrix group the exponential map is simply the matrix exponential

$$X \mapsto \sum_{n=0}^{\infty} \frac{X^n}{n!}.$$

We may summarise some theorems about the exponential map and homomorphisms between Lie groups by requiring that the following diagram commutes

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{df_e} & \mathfrak{h} \\ \exp_G \downarrow & & \downarrow \exp_H \\ G & \xrightarrow{F} & H \end{array}$$

Proposition 1. The exponential map is a diffeomorphism between a neighbourhood U of 0 around the identity in $\mathfrak{g}$ and the neighbourhood of the identity $\exp(U) \subset G$.

The groups $SO(3)$ and $SU(2)$ share a Lie algebra $\mathfrak{su}(2)$, so we do not have a unique association of Lie groups to Lie algebras. We can however recover a very useful correspondence:

Theorem 2.4.1. Among all connected Lie groups G with Lie algebra $\mathfrak{g}$ there exists one unique simply connected Lie group $\tilde{G}$.

Example 2.4.1. Consider $\mathbb{R}$ as a Lie algebra with Lie bracket $[x, y] = 0$ for all x, y . This is the Lie algebra for both $U(1)$ and $\mathbb{R}$ as a Lie group, however only $\mathbb{R}$ is simply connected.

These theorems reduce much of the study of Lie groups to the study of simply connected Lie groups.

Chapter 3

Representation Theory of Lie Groups

"Everything is Representation Theory".

Israel Gel'fand

A representation of a group can be considered a 'linearisation' of the group that preserves the structure of the group.

Definition 3.0.1 (Representation). A representation of a group G is a pair (ρ, V) where V is a vector space and ρ defines an action of G on V such that $\rho(g_1 g_2) = \rho(g_1) \rho(g_2)$.

We are predominantly interested in unitary representations where the operators $\rho(g)^* = \rho(g)^{-1}$ and V is a Hilbert space. This focus stems from quantum mechanics where unitary operators are probability preserving transformations of a state vector.

Definition 3.0.2 (Irreducibility). A representation (ρ, H) is irreducible if the only closed, invariant subspaces in H under the action of ρ are $\{0\}$ and H .

The irreducible representations are the atomic components from which (given some caveats discussed below) any representation is composed.

Definition 3.0.3 (Equivalence). Two representations (ρ, H_1) and (φ, H_2) are called equivalent if there exists an intertwining map $A \in \text{Hom}(H_1, H_2)$ such that

$$A\rho(g) = \varphi(g)A$$

for all $h \in H_1$

Equivalently we may write: two representations are equivalent if there exists a continuous and invertible map $\alpha : H_1 \rightarrow H_2$ that commutes with the action of the group.

$$\alpha(g \cdot h) = g \cdot \alpha(h),$$

where the action on the left hand side is given by $\rho_1(g)$ and the action on the right hand side by $\rho_2(g)$.

Proposition 2. If ρ and φ are equivalent representations then ρ being irreducible implies φ is irreducible.

Irreducible representations can be characterised in the following way:

Theorem 3.0.1 (Schur's Lemma). [9, Proposition 1.5]

A unitary representation Φ of a topological group G on a Hilbert space V is irreducible if and only if the only bounded linear operators on V commuting with all $\Phi(g)$, $g \in G$, are the scalar operators.

Example 3.0.1. Consider the representation of the real line $(\pi, L^2(\mathbb{R}))$ where

$$\pi_a : f(x) \mapsto f(x + a)$$

Clearly

$$\pi_{a+b}f(x) = f(x + a + b) = (\pi_a \circ \pi_b)f(x)$$

so π is a homomorphism.

Consider another representation of the real line $(M, L^2(\mathbb{R}))$

$$M_a : f(x) \mapsto e^{-2\pi i a y} f(x)$$

is also a homomorphism by properties of the exponential.

The Fourier transform $\mathcal{F}$ is a unitary isomorphism on the Hilbert space $L^2(\mathbb{R})$.

$$\begin{aligned} \mathcal{F}(\pi_a(f)) &= \mathcal{F}(f(x + a)) \\ &= \int f(x + a) e^{-2\pi i (x+a)y} dx \\ &= \int f(x + a) e^{-2\pi i (a)y} e^{-2\pi i (x)y} dx \\ &= e^{-2\pi i a y} \hat{f}(y). \end{aligned}$$

Then see,

$$M_a \mathcal{F}(f) = e^{-2\pi i a y} \hat{f}(y).$$

Here we have shown that the Fourier transform $\mathcal{F}$ is an intertwining operator for the action of translation on the Hilbert space $L^2(\mathbb{R})$. Furthermore we can now easily show that π is a reducible representation by noting that the subspace

$$\{e^{-2\pi i a y} \hat{f}(y) \mid a \in \mathbb{R}\} \subset L^2(\mathbb{R}),$$

is invariant under the action of $\mathbb{R}$.

Example 3.0.2. *The Laplacian*

$$\sum_{i=1}^n \frac{\partial^2}{\partial x_i^2},$$

is the intertwining operator for the Euclidean motion group $E(n)$ acting on $L^2(\mathbb{R}^n)$ in the natural way. This fact is commonly stated as the invariance of the Laplacian under orthogonal transformations.

It is important to note that when we refer to a representation we are almost always referring to an equivalence class of representations. The correspondence between representations of the group and the algebra is as follows:

Theorem 3.0.2 ([18, Theorems 3.28]). *Given any representation (ρ, V) of a simply connected Lie group G , there exists a unique representation $d\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ of the Lie algebra $\text{Lie}(G)$.*

3.1 Representations of complex semisimple algebras

The representation theory of semisimple complex Lie algebras developed by Cartan and Weyl is fundamental in the broader study of the representation theory of Lie groups.

Reducibility

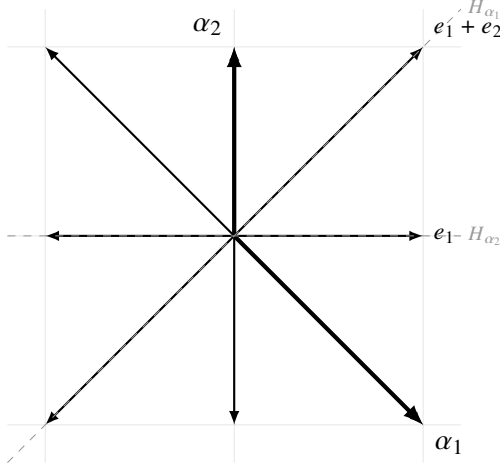
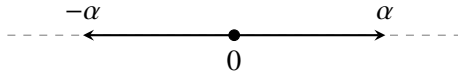
For finite dimensional representations of complex semisimple groups, we have the following important theorem

Theorem 3.1.1 (Weyl's theorem). *Given a semisimple Lie algebra $\mathfrak{g}$ every finite dimensional module (representation) decomposes as a direct sum of simple modules (irreducible representations).*

The infinite dimensional representations require *direct integrals*. The full description of direct integrals of Hilbert spaces is beyond the current scope, we refer the reader to [29, §8.1] for a full treatment of the theory of direct integrals. Suffice to say that the direct integral is the continuous counterpart of the direct sum. The following theorem then provides an infinite dimensional version of 3.1.1:

Theorem 3.1.2 (Gel'fand -Raikov, [3, Appendix 4, §2.4, Theorem 5]). *Any unitary representation of a locally compact group in a separable Hilbert space can be written as a direct integral of irreducible representations.*

This decomposition is unique only for groups of *Type I* in the sense of Von-Neumann. Once again we refer the reader to [29, §5.1] for an in depth discussion. For this work it is enough to know that groups of type one have a 'well-behaved' representation theory and that all semisimple Lie groups are type 1.


 Figure 3.1: B_2 root system.

 Figure 3.2: A_1 root system

Thus the irreducible representations are of central interest as they compose all other representations. Using the *theorem of highest weight*, we are able to classify all finite dimensional irreducible representations. To state the theorem of highest weight we must first provide some definitions.

Definition 3.1.1 (Cartan Subalgebra,[18, §7.2]). Given a complex semisimple Lie algebra $\mathfrak{g}$, the Cartan subalgebra is the complex subspace $\mathfrak{h}$ where

- For all $H_1, H_2 \in \mathfrak{h}$, $[H_1, H_2] = 0$
- Given $X \in \mathfrak{g}$ such that $[H, X] = 0$ for all $H \in \mathfrak{h}$, then $X \in \mathfrak{h}$. Equivalently $\mathfrak{h}$ is a maximal abelian subalgebra.
- Given $H \in \mathfrak{h}$, $\text{ad}H$ is diagonalisable.

Weights

A weight is a linear functional $\lambda \in \mathfrak{h}^*$ where $\mathfrak{h}$ is a Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$. Equivalently one may consider a weight λ as a character of $\mathfrak{h}$. As $\mathfrak{h}$ is abelian see that for $X, Y \in \mathfrak{h}$, $[X, Y] = 0$, so

$$\lambda([X, Y]) = 0 = ([\lambda(X), \lambda(Y)]),$$

thus λ preserves the commutator bracket and defines a homomorphism. As $\mathbb{C}$ is abelian we conclude that a weight λ must vanish on $[\mathfrak{h}, \mathfrak{h}]$. Given a representation $\rho : \mathfrak{g} \rightarrow \text{GL}(V)$ and a weight $\lambda \in \mathfrak{h}^*$ we can form the weight space

$$V_\lambda = \{v \in V \mid \forall H \in \mathfrak{h}, \quad \rho(H) \cdot v = \lambda(H) \cdot v\},$$

V_λ consists of vectors such that the action of any element in $\mathfrak{h}$ is given by scaling by a factor $\lambda(H)$. The functionals λ are then called the weights of ρ .

Roots

Definition 3.1.2. An element α of the dual $\mathfrak{h}^*$ of the Cartan subalgebra $\mathfrak{h}$ is called a root of a Lie algebra $\mathfrak{g}$ when there exists an eigenvector $X \in \mathfrak{g}$ such that

$$\text{ad}_H(X) = \alpha(H)X,$$

i.e. $\alpha(H)$ is an eigenvalue for all $H \in \mathfrak{h}$.

When $\mathfrak{g}$ is defined over $\mathbb{C}$, then by the bilinearity of the Lie bracket, $\text{ad}_H : \mathfrak{g} \rightarrow \mathfrak{g}$ is a linear operator, further $[\text{ad}_{H_1}, \text{ad}_{H_2}] = 0$. so the operators commute and are semisimple. It follows that

$$\mathfrak{g} = \mathfrak{h} + \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha.$$

Positive Roots simple roots and ordering

Given root system Φ we can make a choice of positive roots Φ_+ by taking a hyper plane in the root diagram that does not intersect a root, and choosing the positive roots to be all those contained on one side of the hyperplane. The simple roots often denoted Δ are elements of Φ_+ that cannot be written as the sum of two elements in Φ_+ . The simple roots form a basis for the root space and any root can be written as an integer linear combination of simple roots. In the B_2 example in figure 3.1, the simple roots are written as α_1 and α_2 . We can define a partial ordering on the root space and by extension the entire weight space $\mathfrak{h}^*$ as the simple roots span the dual Cartan subalgebra. Given two weights $\lambda, \mu \in \mathfrak{h}^*$, we define the order $\lambda \preceq \mu$ to mean that the difference $\mu - \lambda$ can be written as a linear combination of simple roots with non-negative integer coefficients:

$$\mu - \lambda = \sum_{\alpha_i \in \Delta} c_i \alpha_i \quad \text{where } c_i \in \mathbb{Z}_{\geq 0}.$$

A weight λ of a representation V is called a highest weight if $\mu \preceq \lambda$ for every other weight μ of V .

Dominant integral weights

Definition 3.1.3. An element $\lambda \in \mathfrak{h}^*$ is an integral weight if its projection onto every simple root $\alpha \in \Delta$ yields an integer:

$$2 \frac{\langle \lambda, \alpha \rangle}{\langle \alpha, \alpha \rangle} \in \mathbb{Z}$$

An integral weight λ is then called dominant if these inner products are strictly non-negative for all positive roots (which is guaranteed if it holds for all simple roots):

$$2 \frac{\langle \lambda, \alpha \rangle}{\langle \alpha, \alpha \rangle} \geq 0 \quad \forall \alpha \in \Delta$$

Combining these two conditions, a dominant integral weight is one where $2 \frac{\langle \lambda, \alpha \rangle}{\langle \alpha, \alpha \rangle} \in \mathbb{Z}_{\geq 0}$.

We may then state the theorem of highest weight:

Theorem 3.1.3 (Highest Weight, [18, Theorems 9.4, 9.5]). *Given a semisimple Lie algebra $\mathfrak{g}$*

- *Every irreducible, finite-dimensional representation of $\mathfrak{g}$ has a highest weight*
- *Finite-dimensional irreducible representations of $\mathfrak{g}$ with the same highest weight are equivalent.*
- *If π, V is an irreducible, finite-dimensional representation of $\mathfrak{g}$ with highest weight μ , then μ is dominant integral.*
- *If μ is a dominant, integral weight then there exists an irreducible, finite-dimensional representation of $\mathfrak{g}$ with highest weight μ .*

Characters

The character of a representation π is defined as follows:

$$\chi_\pi(g) = \text{tr } \pi(g).$$

However for a large class of operators including infinite dimensional representations, the trace is ill-defined so we must generalise the definition of character.

For infinite dimensional representations over a Hilbert space H with basis $\{e_k\}_{k=1}^\infty$,

$$\text{tr } \pi(g) = \sum_{k=1}^{\infty} \langle \pi(g)e_k, e_k \rangle$$

this definition obviously coincides with the usual finite dimensional definition.

The above sum does not necessarily converge for arbitrary infinite dimensional unitary representations in this case we say that the operators are not of trace class, as such the notion

of a generalised character is introduced. We begin by assuming a class of sufficiently nice test densities on the group G , such that

$$\pi(\mu) = \int_G \pi(g) dg$$

is of trace class. The generalised character χ_π is then defined as a linear functional on the space of test densities denoted by μ , where the action of χ_π on a test density μ is given by

$$\langle \chi_\pi, \mu \rangle = \text{tr } \pi(\mu)$$

I.M.Gel'fand defines generalised characters of Lie groups as follows [20, §11.3]: Consider the smooth continuous functions that vanish at infinity $C_0^\infty(G)$.

1. The space $C_0^\infty(G)$ is invariant under left and right translations on G .
2. The representation T of the group G generates a representation $\tilde{T}$ of the algebra $C_0^\infty(G)$ and can be constructed uniquely from this representation¹
3. The operators $T(\phi)$ for $\phi \in C_0^\infty(G)$ are of trace class.

The character of the representation T is then defined as the linear functional χ in the dual of $C_0^\infty(G)$ where

$$\langle \chi, \phi \rangle = \text{tr } \tilde{T}(\phi) = \sum_{n=1}^{\infty} \langle \tilde{T}(\phi) e_k, e_k \rangle$$

χ is a *distribution* on G . This formula holds for semi-simple Lie groups, and for type 1 solvable groups, but can fail in the case of non-type 1 groups. Characters are of utmost importance to representation theory namely due to the theorem:

Theorem 3.1.4. *An irreducible representation T is defined up to equivalence by its character.*

3.2 Compact Lie groups

Compact Lie groups correspond to complex semisimple Lie algebras in the following way:

Theorem 3.2.1. *Given a simply connected compact Lie group K with Lie algebra $\mathfrak{k}$, the complexification $\mathfrak{k}^\mathbb{C}$ is semisimple. Further, every semisimple Lie algebra has a compact real form which coincides with the Lie algebra of a simply connected compact Lie group.*

Lemma 3.2.2. *The following characterise the irreducible representations of compact groups:*

- All irreducible representations of a compact group are unitary.

¹We may associate a representation T of G with a representation $\tilde{T}$ of $C_0^\infty(G)$. $\phi \mapsto \int_G \phi(g) T(g) dg$ See Folland thems 3.9 and 3.11 for the proof that the above defines a representation of $L^1(G)$

- All irreducible representations of a compact group are finite dimensional.

The first point is a consequence of the finitude of the Haar measure over compact Lie groups:

Lemma 3.2.3 ([10, Part 4, §2.1, Proposition 4.6]). *If Φ is a representation of G on a finite-dimensional V , then V admits a Hermitian inner product such that Φ is unitary.*

Proof. Let $\langle \cdot, \cdot \rangle$ be any Hermitian inner product on V , and define

$$(u, v) = \int_G \langle \Phi(x)u, \Phi(x)v \rangle dx.$$

then Φ is unitary with respect to (u, v) . $\square$

Due to the finite dimensionality of the irreducible representations, compact groups admit a narrowing of Weyl's theorem on complete reducibility: For a compact Lie group G , consider a representation λ on a finite vector space vector space W . Then,

$$W \cong \bigoplus_{\pi \in \hat{G}} V_{\pi}^{\oplus m_{\pi}} \quad (3.1)$$

where $\oplus m_{\pi}$ denotes the multiplicity of the representation in the decomposition.

Lemma 3.2.4 ([10, Part 4, §2.1, Corollary 4.10]). *(Schur orthogonality relations). (a) Let Φ and Φ' be inequivalent irreducible unitary representations of G on finite-dimensional spaces V and V' , respectively, and let the Hermitian inner products be denoted $(\cdot, \cdot)$. Then*

$$\int_G (\Phi(x)u, v) \overline{(\Phi'(x)u', v')} dx = 0 \quad \text{for all } u, v \in V \text{ and } u', v' \in V'.$$

(b) Let Φ be an irreducible unitary representation on a finite-dimensional V , and let the Hermitian inner product be denoted $(\cdot, \cdot)$. Then

$$\int_G (\Phi(x)u_1, v_1) \overline{(\Phi(x)u_2, v_2)} dx = \frac{(u_1, u_2) \overline{(v_1, v_2)}}{\dim V} \quad \text{for } u_1, v_1, u_2, v_2 \in V.$$

The orthogonality relations allow one to uncover the multiplicity of a representation in the decomposition.

The Peter-Weyl theorem 1.0.1 can be considered as the compact Lie group counterpart to the Artin-Wedderburn theorem in finite group representations, asserting the semisimplicity of the regular representation. The Peter-Weyl theorem is sometimes phrased:

Corollary 3.2.4.1. *The space $L^2(G)$ is spanned by the matrix coefficients.*

In the work of Gelfand, we see that matrix coefficients form a natural home for many special functions that had been until this point their own objects of study including the spherical harmonics, exponential functions, Legendre polynomials, theta functions, Bessel functions and other families of orthogonal polynomials.

3.3 Non-semisimple groups

After the achievements of Cartan and Weyl in classifying the unitary dual of all compact Lie groups, the natural development was to look to extend the theory to non-compact groups. One path in this direction was to attempt to classify the unitary representations of non-compact semisimple Lie groups. The irreducible unitary representations of a non-compact semisimple group may be infinite dimensional, in fact:

Example 3.3.1. [9, Corollary 2.3] *Every finite dimensional unitary irreducible representation of $SL(2, \mathbb{R})$ is trivial.*

As the theorem of highest weight only classified finite dimensional representations it is thus insufficient for the study of the unitary dual of non-compact semisimple groups such as $SL_n(\mathbb{C})$. This task was taken up by Harish-Chandra from the 1950s - 1970s, who developed tempered representations, a subset of the full unitary dual which proved sufficient to write a Plancherel formula. The classification of the full unitary dual for semisimple non-compact groups proved to be stubborn, the Langlands classification created a class of 'admissible representations', a superset of the unitary dual, proving which of these are unitary has been reduced to a computational task by Vogan whose work formed the theoretical underpinnings of the *Atlas of Lie groups*.

The alternate path was to consider the counterpart of semisimplicity - solvable groups. Dixmier had been working on the representation theory of nilpotent groups in the late 1950s, this was the environment within which Kirillov first developed the orbit method, developing tools for the representation theory of nilpotent Lie groups. We will study the nilpotent case in depth in chapter 5.

3.4 Induced representations

The theory of induced representations allows us to construct a representation of the whole group given only a representation of a subgroup. Induced representations were first developed by Frobenius for finite groups, George Mackey provided the generalisation of this procedure for locally compact second countable groups. For this work it will suffice to present the geometric formulation of induced representations. Further, we restrict our attention to Lie groups (although similar formulations exist for general locally compact groups) and follow the conventions in [14].

Given a Lie group G and a Lie subgroup H , form the quotient manifold $X = G/H$. There is a natural, transitive G action on this space given by

$$g_0H \mapsto gg_0H.$$

Define the *principal H -bundle* as the fibre bundle with projection

$$\pi : G \rightarrow G/H.$$

The inverse image of the bundle projection is

$$\pi^{-1} : \text{im}(\pi) \rightarrow G,$$

which maps points $x \in G/H$ to *fibres* in the total space, in this case the fibres will be copies of the subgroup H in the form of cosets. Given the principal H -bundle, define the *associated vector bundle* as follows: let $\sigma : H \rightarrow GL(V)$ be a representation of H over the vector space V . Write the product space $G \times V$, we then define the action of H on this product space as

$$h(g, v) = (gh, \sigma(h^{-1})v).$$

Define the equivalence $(g, v) \sim (gh, \sigma(h^{-1})v)$ for all $h \in H$. Quotient out by this equivalence relation writing $(G \times V)/\sim$ or $(G \times_H V)$. Consider this as the total space and G/H as the base space, write the bundle projection

$$\begin{aligned} \pi : (G \times V)/\sim &\rightarrow G/H \\ [g, v]_{\sim} &\mapsto gH, \end{aligned}$$

where $[g, v]_{\sim}$ denotes the equivalence class of (g, v) . The inverse image of an element gH will be the orbit $O(g_0) = \{h(g_0) \mid h \in H\} = \{(g_0h) \mid h \in H\}$. The fibres of the bundle projection are copies of the entire vector space V indexed by H orbits in G . Finally we define a *section* of the vector bundle as a map

$$\begin{aligned} s : G/H &\rightarrow (G \times V)/\sim \\ gH &\mapsto [g_0, v_0]_{\sim} \end{aligned}$$

such that $\pi \circ s = \text{Id}$ i.e. $(\pi \circ s)(g_0H) = g_0H$. See that $\pi(s(g_0H)) = g_0H \implies \pi(x) = g_0H$ so x = the fibre above g_0 , therefore s will map $g_0H \rightarrow [g_0, v_0]_{\sim}$. We then write the space of sections as $\Gamma((G \times V)/\sim) = \{s : G/H \rightarrow (G \times V)/\sim \mid v \in V\}$.

Example 3.4.1. Let G be a Lie group with Lie subgroup H . When $V = \mathbb{C}$ and σ is the trivial rep, the associated bundle is the trivial bundle $G/H \times \mathbb{C}$ and the space of sections is all maps $G/H \rightarrow \mathbb{C}$.

The space $(G \times V)/\sim$ is a homogeneous vector bundle and so admits a natural G action

$$(g_0, v) \mapsto (gg_0, v),$$

this induces a G action on $\Gamma((G \times V)/\sim)$, where for $f \in \Gamma((G \times V)/\sim)$

$$f(x) \mapsto f(g^{-1}x),$$

as the space of sections is a vector space, we can identify this action with a linear operator $U(g)$. The map

$$g \mapsto U(g),$$

is a representation of G . In this way, we have produced a representation of the whole group given only a representation of a subgroup H , this representation is called the *induced representation*.

Unitarity

If the inducing representation ρ is unitary, then the induced representation is unitary as the space G/H admits a G invariant measure μ such that $\int_{G/H} f(x) d\mu = \int_{G/H} f(g \cdot x) d\mu$. Defining then the inner product

$$\langle f_1, f_2 \rangle = \int_{G/H} \langle f_1(x), f_2(x) \rangle_V d\mu(xH),$$

see then that

$$\langle g \cdot f_1, g \cdot f_2 \rangle = \int_{G/H} \langle f_1(g^{-1}x), f_2(g^{-1}x) \rangle_V d\mu(xH) = \int_{G/H} \langle f_1(x), f_2(x) \rangle_V d\mu(xH)$$

so the action of G is unitary. This is only true for unimodular groups, e.g. compact, semi-simple, nilpotent. In these cases, the modular functions satisfy $\Delta_G|_H = \Delta_H = 1$. Consequently, the quotient space G/H is guaranteed to admit a strictly G -invariant measure μ , ensuring that our inner product is well-defined and the induced representation is unitary.

Equivariant functions

When the inducing representation ρ is one dimensional, the associated vector bundle is a *line bundle* L_ρ . The space of sections $\Gamma(L_\rho)$ of the associated line bundle is equivalent to the space of equivariant functions on the group

$$\{f : G \rightarrow V \mid f(gh) = \rho(h^{-1})f(g)\},$$

under the association $f \leftrightarrow s$ where $s(gH) = (g, f(g))$.

Constructing representations

To explicitly construct the induced representation we follow the method laid out by Kirillov in [3, Appendix 5, §2.2] Kirillov uses somewhat non-standard terminology however the process is similar to that of Mackey or Wigner.

We first consider the homogeneous right coset space $X = H/G$ and the principle bundle structure

$$\begin{aligned} \pi : G &\rightarrow X \\ g &\mapsto Hg, \end{aligned}$$

with sections

$$s : X \rightarrow G.$$

Notice then that for all $g \in G$

$$g = hs(x),$$

as $s(x)$ will equal the identity over $x \in H$ so the multiplication of h accounts for this. Kirillov then writes the cocycle equation, which he coins the 'master equation'

$$s(x)g = h(x, g)s(y) \quad (3.2)$$

where $x \in X, g \in G, y = x \cdot g$ and $h(x, g) : X \times G \rightarrow H$ is defined as to satisfy the master equation.

Lemma 3.4.1. *The map $h(x, g)$ will satisfy:*

$$h(x, g_1 g_2) = h(x, g_1)h(x \cdot g_1, g_2)$$

Proof.

$$\begin{aligned} h(x, g_1) \cdot h(xg_1, g_2) &= s(Hg)g_1 s(Hgg_1)^{-1} \cdot s(Hgg_1)g_2 s(Hh(g_1 g_2))^{-1} \\ &= s(x)g_1 g_2 s(y)^{-1} = h(x, g_1 g_2) \end{aligned}$$

□

Further, we can identify $\phi : G \rightarrow V$ with $f : X \rightarrow V$ via

$$f(x) = \phi(s(x)),$$

and the induced representation will be of the form

$$(\text{Ind}_H^G \rho)(g)(f)(x) = A(x, g)f(x \cdot g), \quad (3.3)$$

where $A(x, g) = \rho(h(x, g))$.

One can see that we have significantly simplified the construction of the induced representation where we have encoded the equivariance property of ϕ through the composition with $s : X \rightarrow G$.

Example 3.4.2. *Let $G = SU(2)$ with subgroup $T = U(1)$, written in matrix form:*

$$T = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}$$

each $t \in T$ can then be written as a function of θ so we make the identification $t \leftrightarrow \theta$. Let

$$\begin{aligned} \rho : T &\rightarrow \mathbb{C} \\ \theta &\mapsto e^{in\theta} \end{aligned}$$

for some $n \in \mathbb{Z}$. We then form the product space $G \times \mathbb{C}$ where the action of T is defined as

$$t(g, \lambda) = (gt, \rho(t^{-1})\lambda).$$

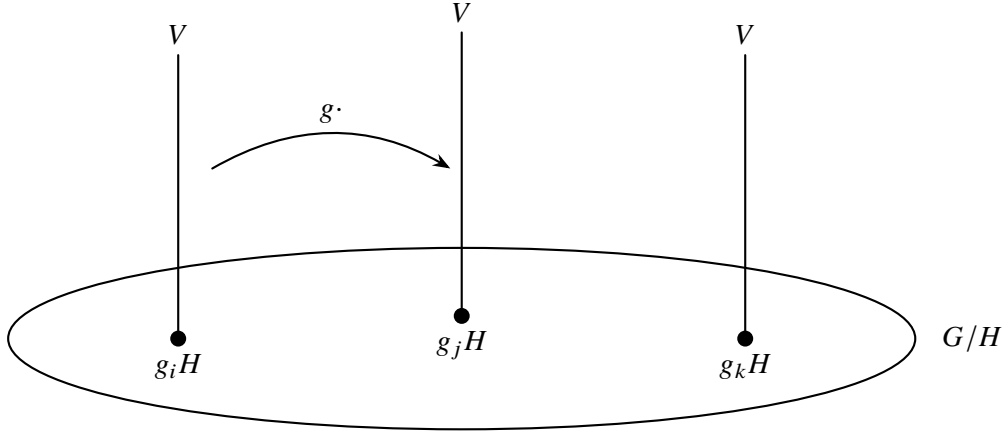


Figure 3.3: The induced representation $\text{Ind}_H^G(\rho)$, G acts on the fibres.

Define an equivalence relation $\sim$ with respect to the action of T :

$$(g, \lambda) \sim t(g, \lambda),$$

and quotient out by this relation:

$$L_\rho = (G \times \mathbb{C}) / \sim.$$

We can then define the induced representation

$$(\text{Ind}_T^G \rho) = \{f \in \Gamma(E_{\mathbb{C}})\}$$

where G acts by

$$(\text{Ind}_T^G \rho)(g)f = f(g^{-1}x)$$

Holomorphic line bundles

Continuing now the example of $SU(2)$. We will see in chapter 7 that the orbit Ω_λ admits a complex structure. The associated line bundle $L_\rho = SU(2) \times \mathbb{C} / \sim$ inherits the projection from the principal T bundle G/T where

$$\pi((g, \lambda)) = g/T,$$

and can be shown to be a holomorphic bundle in the following way: As $\pi : E_{\mathbb{C}} \rightarrow \Omega_\lambda \cong SU(2)/T$ is a complex line bundle, for any open neighbourhood U around a point $p \in \Omega_\lambda$ then there is a topological isomorphism (homeomorphism)

$$\varphi : U \times \mathbb{C} \rightarrow \pi^{-1}(U)$$

such that

- $(\pi \circ \varphi)(g, \lambda) = g$ for all $\lambda \in \mathbb{C}$
- the map $\lambda \mapsto \varphi(g, \lambda)$ is a linear isomorphism between $\mathbb{C}$ and $\pi^{-1}(\{g\})$.

The complex line bundle is called holomorphic if the maps called *local trivialisations*:

$$\varphi : U \times \mathbb{C} \rightarrow \pi^{-1}(U).$$

where U is an open subset of $SU(2)$ are holomorphic.

The holomorphic sections of a holomorphic line bundle are then the holomorphic mappings

$$s : \Omega \rightarrow L_\rho,$$

such that $\pi \circ s = id$.

Chapter 4

Coadjoint Orbits

Here we collect some definitions and theorems surrounding the central object of our study, namely coadjoint orbits. We recommend that the reader gains a general familiarity with these structures before continuing, however the details can be left to a second pass after the motivation has been properly established in the representation theory of nilpotent groups.

4.1 Preliminary definitions

Adjoint representation

The adjoint representation of a Lie group takes the Lie algebra as the vector space over which the representation acts:

$$\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g}).$$

If G is a matrix group, then

$$\text{Ad}(g)X = gXg^{-1}.$$

The differential of this representation:

$$(\text{Ad})_{*,e} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g}),$$

where $\mathfrak{gl}(\mathfrak{g})$ is the set of matrices acting on $\mathfrak{g}$, is the adjoint representation of the Lie algebra, $\text{ad} := (\text{Ad})_{*,e}$.

$$\begin{aligned} \text{ad}(X)Y &= \left. \frac{d}{dt} \text{Ad}(\exp tX) \cdot Y \right|_{t=0} = \left. \frac{d}{dt} (\exp tX) \cdot Y \cdot (\exp tX)^{-1} \right|_{t=0} \\ &= \left. X \exp(tX) \cdot Y \cdot \exp(-tX) - \exp(tX) \cdot Y \cdot X \exp(-tX) \right|_{t=0} \\ &= XY - YX = [X, Y]. \end{aligned}$$

Contragredient representation

Given any representation ρ of a group G on a vector space V over a field $\mathbb{F}$, we may form the dual space V^* , the vector space of linear functions $f : V \rightarrow \mathbb{F}$. It is natural then to define a representation $\rho^* : G \rightarrow GL(V^*)$. We call ρ^* the contragredient or dual representation to ρ and define it:

$$\rho^*(g) := \rho(g^{-1})^*. \quad (4.1)$$

Where $\rho(g)^*$ is the adjoint operator to $\rho(g)$. If $\langle \cdot, \cdot \rangle$ is a bilinear map $\langle \cdot, \cdot \rangle : V^* \times V \rightarrow \mathbb{F}$. Write $\langle f, v \rangle$ to denote the evaluation of $f \in V^*$ at $v \in V$, then $\rho(g)^*$ is the operator such that

$$\langle \rho(g)^* f, v \rangle = \langle f, \rho(g)v \rangle.$$

Lemma 4.1.1. *The function defined in equation 4.1 is a representation of G on V^* .*

Proof.

$$\rho^* : G \rightarrow GL(V^*), \quad g \mapsto \rho(g^{-1})^* \quad (4.2)$$

Notice, $\rho^*(gh) = \rho(h^{-1}g^{-1})^*$ then,

$$\langle \rho(h^{-1}g^{-1})^* f, v \rangle = \langle f, \rho(h^{-1}g^{-1})v \rangle = \langle f, \rho(h^{-1})\rho(g^{-1})v \rangle$$

to see,

$$\langle f, \rho(h^{-1})\rho(g^{-1})v \rangle = \langle \rho(g^{-1})^* \rho(h^{-1})^* f, v \rangle.$$

then

$$\rho(g^{-1})^* \rho(h^{-1})^* = \rho^*(g)\rho^*(h).$$

Therefore ρ^* is a homomorphism, and a representation of the group G on the dual V^* . $\square$

Contragredient representation of Adjoint representation

Given the adjoint representation of a Lie group G on its Lie algebra $\mathfrak{g}$, we may then find its contragredient representation, $\text{Ad}^* : G \rightarrow GL(\mathfrak{g}^*)$. From the definition in eq.4.1,

$$\text{Ad}^*(g) := \text{Ad}(g^{-1})^*.$$

Recalling the evaluation map, $\langle F, X \rangle$, for $F \in \mathfrak{g}^*$, $X \in \mathfrak{g}$, the contragredient of the adjoint representation is then the operator that satisfies the following,

$$\langle \text{Ad}^*(g)F, X \rangle = \langle F, \text{Ad}(g^{-1})X \rangle$$

As in [3, 1.1] we use the abbreviated notation:

$$\langle K(g)F, X \rangle := \langle \text{Ad}^*(g)F, X \rangle = \langle F, \text{Ad}(g^{-1})X \rangle,$$

for the remainder of this work, and call $K : G \rightarrow GL(\mathfrak{g}^*)$ the *Coadjoint representation* of a Lie group G . The coadjoint action on a matrix group is defined

$$K(g)F(X) = F(\text{Ad}(g^{-1})X) = F(gXg^{-1}), \quad (4.3)$$

so, $K(g)F = F \circ \text{Ad}(g^{-1})$.

The differential of the coadjoint representation is: $K_* : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g}^*)$,

$$\langle K_*(X)F, Y \rangle = \langle \text{ad}^*(X)F, Y \rangle = \langle F, \text{ad}(-X)Y \rangle = \langle F, [Y, X] \rangle. \quad (4.4)$$

Calculation of duals and coadjoint action

In order to calculate the coadjoint orbits for a finite dimensional Lie group we translate the abstract definitions into linear algebraic statements. Given any n dimensional Lie algebra $\mathfrak{g}$, the existence of a faithful representation on $GL(V)$ allows us choose a basis and consider $\mathfrak{g} \subset \text{Mat}_n(\mathbb{C})$ the space of $n \times n$ matrices with complex coefficients. In order to explicitly realise the dual $\mathfrak{g}$ we use the theorem:

Theorem 4.1.2. ¹ Given vector spaces W, V with $W \subset V$ and a bilinear form $(\cdot, \cdot)$,

$$(V/W)^* \cong W^\perp$$

where $(V/W)^*$ denotes the dual, and the orthogonal complement $W^\perp$ is taken with respect to the bilinear form $(\cdot, \cdot)$.

Taking $V = \text{Mat}_n(\mathbb{F})$ and $W = \mathfrak{g}^\perp$ gives,

$$(\text{Mat}_n(\mathbb{F})/\mathfrak{g}^\perp)^* \cong \mathfrak{g}$$

taking dual of both sides noting that for finite dimensional V we have $V^{**} \cong V$,

$$\text{Mat}_n(\mathbb{F})/\mathfrak{g}^\perp \cong \mathfrak{g}^*.$$

The space $\text{Mat}_n(\mathbb{F})/\mathfrak{g}^\perp$ may be associated with a subspace U of $\text{Mat}_n(\mathbb{F})$ such that $\text{Mat}_n(\mathbb{F}) = U \oplus \mathfrak{g}^\perp$.

The algebra $\text{Mat}_n(\mathbb{F})$ admits a non-degenerate bilinear form

$$(A, B) = \text{tr}(AB)$$

the action of an element of the dual on the Lie algebra is calculated using this form. The coadjoint representation $K = \text{Ad}^*$ is then realised as

$$(\text{Ad}^*(g)F, X) = (F, \text{Ad}(g^{-1})X) = (F, g^{-1}Xg) = \text{tr}(Fg^{-1}Xg).$$

In order to maintain agreement with the abstract definition, we use the conjugate invariance of the trace to write

$$\text{tr}(Fg^{-1}Xg) = \text{tr}(gFg^{-1}X)$$

¹This theorem is usually stated in terms of the annihilator W° of W , we use the natural isomorphism $V \cong V^*$ to associate W° with $W^\perp$

so we may write the coadjoint action as

$$K(g) : F \mapsto gFg^{-1}$$

where

$$(K(g)F, X) = (F, \text{Ad}(g^{-1})X).$$

This map needs a corrective projection, as $F \mapsto gFg^{-1}$ may map F to a point outside $V = \text{Mat}_n(\mathbb{F})/\mathfrak{g}^\perp$. Any point outside V will define a map that is trivial on $\mathfrak{g}^2$, so we write the projection:

$$\begin{aligned} p_V : \text{Mat}_n(\mathbb{F}) &\rightarrow V \\ M &\mapsto Mg^\perp. \end{aligned}$$

Any $M \in \text{Mat}_n(\mathbb{F})$ can be written $M = V + X$ where $V \in V$ and $X \in \mathfrak{g}^\perp$. Letting M act on an element H of the algebra:

$$(M, H) = \text{tr}(MH) = \text{tr}((V + X)H) = \text{tr}(VH + XH) = \text{tr}(VH)$$

$(X, \cdot) = 0$ on $\mathfrak{g}$. The coadjoint action is then,

$$K(g) : F \mapsto p_V(gFg^{-1}).$$

Orbits

Given a group action, it is natural to ask about its orbits in the range. The orbits of the adjoint action of the group G on its Lie algebra $\mathfrak{g}$ are written

$$O(X) = \{\text{Ad}(g)X \mid \forall g \in G\} \quad X \in \mathfrak{g}, g \in G.$$

These are subsets of $\mathfrak{g}$. In the case of matrix lie groups we may write the orbits explicitly as

$$O(X) = \{g^{-1}Xg \mid \forall g \in G\} \quad X \in \mathfrak{g}, g \in G.$$

Notice that for $A, B \in O(X)$, $A = g^{-1}Xg$, $B = h^{-1}Xh$, so $gAg^{-1} = hBh^{-1} = X$, so A and B are similar matrices. These orbits partition $\mathfrak{g}$ into conjugacy classes. The coadjoint orbits are defined similarly,

$$O(X) = \{K(g)F \mid \forall g \in G\} \quad F \in \mathfrak{g}^*, g \in G.$$

²Recall that $\mathfrak{g}^\perp \cong \mathfrak{g}^\circ$ so all the maps in $\text{Mat}_n(\mathbb{F})/\mathfrak{g}^\perp$ are non-trivial on $\mathfrak{g}$

Stabiliser of functional

Let F be an element in $\mathfrak{g}^*$. The stabiliser of F under the coadjoint action is defined to be

$$\text{Stab}(F) = \{g \in G \mid K(g)F = F\}$$

As the coadjoint action is smooth, the stabiliser is a Lie subgroup of G . Let $\text{stab}(F)$ be the Lie subalgebra such that $\exp \text{stab}(F) = \text{Stab}(F)$. Then for $X \in \text{stab}(F)$,

$$K_*(X)F = \left. \frac{d}{dt} \right|_{t=0} K(\exp tX)F = \left. \frac{d}{dt} \right|_{t=0} F = 0$$

where K_* is the differential of the coadjoint representation. Then

$$\langle K_*(X)F, Y \rangle = \langle F, \text{ad}(-X)Y \rangle = 0 \quad \forall Y \in \mathfrak{g}$$

So write $\text{stab}(F)$ as

$$\text{stab}(F) = \{X \in \mathfrak{g} \mid F([X, Y]) = 0 \text{ for all } Y \in \mathfrak{g}\}. \quad (4.5)$$

Write the map

$$\phi : X \mapsto 2\pi i \langle F, X \rangle \quad X \in \text{stab}(F)$$

then

$$\phi([X, Y]) = 2\pi i \langle F, [X, Y] \rangle = 0$$

and

$$[\phi(X), \phi(Y)] = 0$$

therefore ϕ is a one dimensional Lie algebra representation of the subalgebra $\text{stab}(F)$. Define another map

$$\rho : \text{Stab}(F) \rightarrow \mathbb{C} \quad (4.6)$$

$$\exp X \mapsto e^{2\pi i \langle F, X \rangle} \quad (4.7)$$

For G simply connected, ρ forms a representation of $\text{Stab}(F)$ as in 3.0.2.

4.2 Canonical 2-form on a coadjoint orbit

At the heart of the machinery of the orbit method lies the fact that the coadjoint orbits Ω of a Lie group G are naturally realised as submanifolds of the dual Lie algebra $\mathfrak{g}^*$, and are canonically equipped with a two-form that endows them with the structure of a symplectic manifold.

Differential forms and vector fields

Definition 4.2.1. A k -form ω on a manifold M is a smooth section of the k -th exterior power of the cotangent bundle, $\Lambda^k(T^*M)$.

Consider a manifold M , an open set $U \subset M$ with local coordinates $x_1, \dots, x_n$, and a function $f \in C^\infty(M)$. The tangent space at $m \in U$ is spanned by vectors,

$$\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n}$$

which we consider as differential operators. Let x_i be the i th coordinate in U . A vector field $\hat{X} = \sum \alpha_i \frac{\partial}{\partial x_i}$ evaluated at a point $m \in M$ is written

$$\hat{X}(m) = \sum \alpha_i(m_i) \frac{\partial}{\partial x_i},$$

i.e. a linear combination of basis vectors for the tangent space $T_m M$, so $\hat{X}(m)$ is a point in $T_m M$.

The differential of a function on M is defined:

$$df(\hat{X}(m)) := \hat{X}(m)(f)$$

where $\hat{X}(m)$ is any element of the tangent space $T_x M$ and

$$\hat{X}(m)(f) = \sum \alpha_i(m_i) \frac{\partial}{\partial x_i} f,$$

is interpreted as a differential operator. The differential of the coordinate $x_i \in U$ is given:

$$dx_i(\hat{X}) := \hat{X}(x_i) = \sum_{j=1}^n \alpha_j \frac{\partial}{\partial x_j} x_i = \alpha_i \frac{\partial}{\partial x_i} x_i = \alpha_i.$$

As $dx_i \frac{\partial}{\partial x_j} = \delta_{i,j}$, the dx_i form a dual basis for the tangent space in the usual way. So $T_m^* M$ has local basis $dx_1, \dots, dx_n$.

Given a k -form α and an l form β , define

$$\alpha \otimes \beta = \alpha(x_1, \dots, x_k) \beta(x_{k+1}, \dots, x_{k+l})$$

The wedge product is then defined by

$$\alpha \wedge \beta = AS(\alpha \otimes \beta)$$

where AS is the antisymmetrisation map:

$$AS(\alpha)(x_1, \dots, x_k) := \sum_{\sigma \in S_k} \text{sgn}(\sigma) \alpha(x_{\sigma(1)}, \dots, x_{\sigma(k)})$$

and S_k is the symmetric group on k elements.

Example 4.2.1. Given 1-forms α and β ,

$$(\alpha \otimes \beta)(\hat{X}_1, \hat{X}_2) = \alpha(\hat{X}_1)\beta(\hat{X}_2)$$

then

$$\begin{aligned} \alpha \wedge \beta(\hat{X}_1, \hat{X}_2) &= \sum_{\sigma \in S_2} \text{sgn}(\sigma)(\alpha \otimes \beta)(\hat{X}_{\sigma(1)}, \hat{X}_{\sigma(2)}) \\ &= (\alpha \otimes \beta)(\hat{X}_1, \hat{X}_2) - (\alpha \otimes \beta)(\hat{X}_2, \hat{X}_1) \\ &= \alpha(\hat{X}_1)\beta(\hat{X}_2) - \alpha(\hat{X}_2)\beta(\hat{X}_1) \end{aligned}$$

Remark. The domains of definition can become confusing here. Succinctly, a k form α assigns $x \in M$ to a functional on the tangent space at x , written α_x or $\alpha(x)$. This functional maps a k -tuple of elements from the tangent space $T_x M$ to the underlying field. Often we write $\alpha_x(X_1, \dots, X_k)$ where the X_i are vector fields. Recalling that a vector field is an assignment of $x \in M$ to a point in $T_x M$, $\alpha_x(X_1, \dots, X_k)$ should be interpreted as the functional α_x acting on the tangent vectors $X_i(x) \in T_x M$. When we see $\alpha(X_1, \dots, X_k)$ with no subscript x defined we interpret this as the general functional with domain yet to be specified. See [30, Chapter 21] for more detail.

Example 4.2.2. Taking (v_1, v_2) to be a point in $T_p M \times T_p M$, we evaluate a two form at the point p as $\omega(p) = dx \wedge dy^3$ as,

$$\omega(p)(v_1, v_2) = (dx(v_1)dy(v_2) - dx(v_2)dy(v_1))$$

Here, dx and dy are one forms, that is elements of $T_p M^*$. The term $dx(v_1)$ can be interpreted as

$$dx(v_1) = \sum_{i,j}^n (\beta_j dx_j)(\alpha_i \partial_i),$$

where the ∂_i terms form the basis of the tangent space $T_p M$ and the dx_i terms form the basis for the cotangent space at the same point. Define

$$dx_i(\partial_j) = \delta_{i,j},$$

using this definition we see

$$dx(v_1) = \sum_i^n (\beta_i)(\alpha_i),$$

then

$$\omega(p)(v_1, v_2) = (dx(v_1)dy(v_2) - dx(v_2)dy(v_1)) = \sum_i^n \beta_i \alpha_i \sum_j^n c_j d_j - \sum_j^n c_j d_j \sum_i^n \beta_i \alpha_i.$$

³Note that all two forms can be written as the sum of wedge products of one forms in local coordinates, by the Darboux theorem.

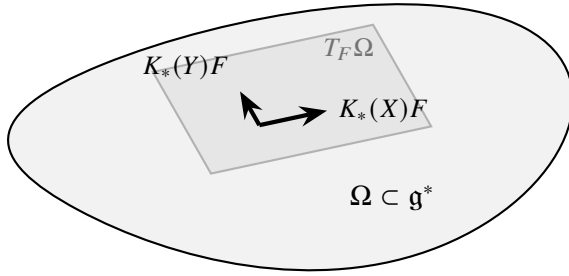


Figure 4.1: Tangent plane to orbit manifold

Kirillov-Kostant-Souriau form

A common example of a symplectic manifold is the cotangent bundle T^*M of a manifold M , which in a canonical way admits a non-degenerate two-form defining the symplectic structure. It is natural then to ask that for a coadjoint orbit we develop a similar construction, to this end, the Kirillov-Kostant-Souriau form is defined as

$$\sigma_\Omega(\beta)(K_*(X)\beta, K_*(Y)\beta) = \langle \beta, [X, Y] \rangle \quad X, Y \in \mathfrak{g}, \quad (4.8)$$

where K_* is the coadjoint representation of the Lie algebra. The KKS form is the explicit link between the algebraic properties of the group and the geometric properties of the orbit.

The KKS form σ_Ω satisfies the following:

Theorem 4.2.1. [34, Theorem 5.1]

1. σ_Ω is preserved by the action of G
2. σ_Ω is closed
3. σ_Ω is non degenerate on the orbit through β

Tangent space to an orbit

An orbit of the group in the dual of the Lie algebra,

$$\Omega_F = \{K(g)F \mid g \in G\},$$

can be considered as a manifold. The tangent space to a point F on the orbit is then

$$T_F \Omega_F = \{K_*(X)F \mid X \in \mathfrak{g}\}.$$

We make this notion precise by writing group G as a principal $Stab(F)$ bundle over the orbit with projection [3, §1.2]

$$\begin{aligned} P_F : G &\rightarrow \Omega \\ g &\mapsto K(g)F, \end{aligned}$$

where the fibre above F is $Stab(F)$. The projection map is a submersion [33, Theorem 4.14], therefore the differential

$$\begin{aligned} (P_F)_* : \mathfrak{g} &\rightarrow T_F\Omega \\ X &\mapsto K_*(g)F \end{aligned}$$

is a surjective map. Write the exact sequence

$$0 \rightarrow stab(F) \hookrightarrow \mathfrak{g} \xrightarrow{(P_F)_*} T_F\Omega \rightarrow 0$$

such that

$$T_F\Omega \cong \mathfrak{g}/stab(F).$$

Symplectic manifolds

A manifold equipped with a closed non degenerate two form is called a *symplectic manifold*. Every coadjoint orbit can be considered a symplectic manifold with respect to the canonical KKS 2-form defined above. In fact a partial converse to this fact holds:

Theorem 4.2.2. [28, Theorem 1.1] *Every G -orbit in $\mathfrak{g}^*$ is a homogeneous symplectic G -manifold and conversely, every homogeneous symplectic G -manifold is locally isomorphic to an orbit in the coadjoint representation of the group G or a central extension of it.*

Volume Forms and measures

A volume form is a nowhere vanishing differential form of 'top-degree', that is given a manifold M of dimension d , a volume form will have degree d . For a symplectic manifold the dimension is always even, $\dim M = 2d$, the canonical KKS 2-form then gives rise to a volume form by considering the d th power $\omega^d := \omega \wedge \cdots \wedge \omega$. This volume form is unique up to factor of a constant multiple.

On an oriented manifold M , a volume form ω gives rise to a measure μ_ω via

$$\mu_\omega(A) = \int_A \omega$$

Chapter 5

Representation Theory of Nilpotent Lie Groups

Here we gather some key theorems due to Kirillov on the representation theory of nilpotent groups and present the Heisenberg group as an example. These theorems mark the first explicit codification of the orbit method. There are significant differences compared with the compact group case, stemming from existence of infinite dimensional irreducible representations, which necessitate the development of some more nuanced analytical tools. The proof of the orbit method for nilpotent groups proceeds by induction on the dimension of the group using the fact that a subgroup of a nilpotent group is nilpotent, the base case is $\dim(G) = 1$ where $G = \mathbb{R}$. The proofs form the content of Kirillov's paper [7] and a brief treatment can be found in [3].

5.1 Unitary dual of nilpotent Lie groups

The two theorems below completely classify the unitary irreducible representations of any simply connected nilpotent Lie group.

Theorem 5.1.1 (Kirillov, Dixmier, [7, Theorem 5.1]). *Every irreducible unitary representation T of a nilpotent Lie group G is induced from a one-dimensional representation of some subgroup.*

Corollary 5.1.1.1. *Every irreducible unitary representation of a connected nilpotent Lie group is either one-dimensional or infinite dimensional.*

Theorem 5.1.2 (Kirillov, [7, Theorem 5.2]). *For every functional f on $\mathfrak{g}$ and every subalgebra $\mathfrak{g}_1$ subordinate to f , denote by $T^{(f, \mathfrak{g}_1)}$ the representation of $\mathfrak{g}$ that is induced by the one-dimensional representation $U(\exp(x)) = e^{i\langle f, x \rangle}$ of the subgroup $G_1 = \exp(\mathfrak{g}_1)$. Then,*

1. *The representation $T^{(f, \mathfrak{g}_1)}$ is irreducible if and only if $\mathfrak{g}_1$ is a subalgebra of maximal dimension subordinate to f .*

2. The irreducible representations $T^{(f, \mathfrak{g}_1)}$ and $T^{(f_2, \mathfrak{g}_2)}$ are equivalent if and only if the functionals f, f_2 lie in the same G orbit under the coadjoint action.

These two theorems admit an elegant geometric summarisation:

Theorem 5.1.3. *The unitary irreducible representations of a simply connected nilpotent Lie group G are in bijection with the coadjoint orbits of G in $\mathfrak{g}^*$.*

This correspondence is the heart of the orbit method.

Theorem 5.1.4 ([19]). *There is a homeomorphism between $\hat{G}$ with the Fell topology and $O(G)$ with the quotient topology.*

5.2 Restriction and Induction

The restriction and induction of representations can be described in terms of coadjoint orbits.

Theorem 5.2.1 ([?]). *Let G be a simply connected nilpotent Lie group, G_0 a closed subgroup and p^* the natural projection of $\mathfrak{g}^*$ onto $\mathfrak{g}_0^*$.*

1. *If T is an irreducible representation of G corresponding to the orbit $\Omega \subset \mathfrak{g}^*$, then the restriction of T to G_0 decomposes into the direct integral of irreducible representations of G_0 corresponding to all orbits in $\mathfrak{g}_0^*$ contained in $p^*(\Omega)$*
2. *If T is an irreducible representation of the subgroup G_0 corresponding to the orbit $\Omega \subset \mathfrak{g}_0^*$ then the representation of G induced by T decomposes into the direct integral of irreducible representations of G corresponding to all the orbits in $\mathfrak{g}^*$ that meet the inverse image $p^{*-1}(\Omega)$*
3. *If T_1 and T_2 are irreducible representations of G corresponding to the orbits Ω_1 and Ω_2 , then their tensor product $T_1 \otimes T_2$ decomposes into the direct integral of the irreducible representations of G corresponding to the orbits in $\mathfrak{g}^*$ that are contained in the arithmetic sum of the orbits Ω_1 and Ω_2 .*

If we consider the homogeneous space $X = G/G_0$, the space of functions $L^2(X)^1$, then part 2 of theorem 5.2.1 provides the decomposition of the left-regular G representation on this space.

Given the subalgebra $\mathfrak{h} \subset \mathfrak{g}$ corresponding to the subgroup $H \subset G$ there is a natural projection

$$\begin{aligned} p : \mathfrak{g}^* &\rightarrow \mathfrak{h}^* \\ F &\mapsto F|_H. \end{aligned}$$

¹This space can be thought of as the space of sections where the action of H is trivial, i.e. given by $h(g, v) = (g, v)$, and $V = \mathbb{C}$ see §3.4 for more detail

We can then write the restriction formula as

$$\text{Res}_H^G \pi_\Omega = \int_{\omega \in p(\Omega)} m(\Omega, \omega) \pi_\omega d\omega. \quad (5.1)$$

In words this says that the representation of G associated with the orbit Ω restricted to the subgroup H is given by the integral (over the projection $p(\Omega)$ to the subalgebra $\mathfrak{h}^*$) of the representation π_ω of H associated to an orbit $\omega \in \mathcal{O}(H)$ multiplied by a multiplicity factor $m(\Omega, \omega)$. This multiplicity factor can also be realised geometrically: write $\Gamma_p \subset \mathfrak{g}^* \times \mathfrak{h}^*$, to be the graph of the projection p . Given coadjoint orbits Ω in $\mathfrak{g}^*$ and $\omega \in \mathfrak{h}^*$ we define $k(\Omega, \omega)$ as

$$k(\Omega, \omega) = \dim(\Gamma_p \cap (\Omega \times \omega)) - \frac{1}{2}(\dim \Omega + \dim \omega)$$

The multiplicity function is then defined

$$m(\Omega, \omega) = \begin{cases} 0 & \text{if } k(\Omega, \omega) < 0 \\ 1 & \text{if } k(\Omega, \omega) = 0 \\ \infty & \text{if } k(\Omega, \omega) > 0 \end{cases}$$

5.3 Characters of representations

The Kirillov character formula calculates the generalised character χ for a nilpotent Lie group in terms of the coadjoint orbits

$$\chi = \int_{\Omega} \tilde{\phi}(f) d\omega(f),$$

where $\phi \in C_0^\infty(G)$, $\tilde{\phi}$ denotes the Fourier transform, and $d\omega(f)$ is the canonical measure on the orbit. This formula is evaluated by pulling it back to the Lie algebra. For a simply connected nilpotent Lie group, the exponential map $\exp : \mathfrak{g} \rightarrow G$ is a diffeomorphism, and its Jacobian $j(X) = 1$. This allows us to identify the Haar measure dx on G with the Lebesgue measure dX on $\mathfrak{g}$. Letting $x = \exp X$ for $X \in \mathfrak{g}$, the Kirillov character formula is written:

$$j(X)\chi(\exp x) = \int_{\mathcal{O}_{\lambda+\delta}} e^{2\pi i(\beta, X)} d\mu_{\lambda+\delta}(\beta). \quad (5.2)$$

This formula holds in the sense of distributions, that is when tested against test functions

$\phi \in C_c^\infty$:

$$\begin{aligned}
& \int \phi(x) \int_{O_{\lambda+\delta}} e^{2\pi i(\beta, X)} d\mu_{\lambda+\delta}(\beta) dx \\
&= \int \phi(x) e^{2\pi i(\beta, X)} \int_{O_{\lambda+\delta}} d\mu_{\lambda+\delta}(\beta) dx \\
&= \int_{O_{\lambda+\delta}} \int \phi(x) e^{2\pi i(\beta, X)} dx d\mu_{\lambda+\delta}(\beta) \\
&= \int_{O_{\lambda+\delta}} \widehat{\phi(x)} d\mu_{\lambda+\delta}(\beta),
\end{aligned}$$

where $\phi \in C_c^\infty$, recovering the original formula as the Jacobian $j(X)$ is equal to 1 for nilpotent groups.

5.4 Construction of irreducible representations

Construction by induction from a polarisation

For nilpotent Lie groups we have seen that every irreducible unitary representation is induced from a one-dimensional representation of a subgroup. Further, we have seen that every irreducible representation is uniquely associated with a coadjoint orbit. Given a coadjoint orbit Ω we can construct a unitary representation by the following method:

- Choose a functional $F \in \Omega$
- Find a subalgebra $\mathfrak{b} \subset \mathfrak{g}$ subordinate to F of maximal dimension
- Define the character of the subgroup B by $\rho(\exp X) = e^{2\pi i\langle F, X \rangle}$
- Choose a section $s : H/B \rightarrow B$ and solve the master equation

$$s(x) \cdot g = h(x, g) \cdot s(x \cdot g)$$

- The induced representation on $L^2(H/B, \mu_s)$ is then given by

$$(\pi_\Omega(\exp X)f)(x) = e^{2\pi i\langle F, X \rangle} f(x \cdot g)$$

where $B = \exp(\mathfrak{b})$ and $e^{2\pi i\langle F, X \rangle}$ is the one-dimensional unitary representation of the subgroup B . We then call π_Ω the irreducible representation associated with the orbit Ω . This representation is independent of the choice of polarisation [3, Chapter 2, pp. 66–69].

Polarisations and subalgebras

The subalgebra of maximal dimension condition may initially appear unmotivated; it is required so that we can define a character of the subgroup $\exp \mathfrak{b}$ by $\rho(\exp X) = e^{2\pi i \langle f, X \rangle}$ where $X \in \mathfrak{b}$.

Firstly, we consider a one dimensional unitary representation U of G , as in theorem 3.0.2 U gives rise to a Lie algebra representation of the form

$$x \mapsto i\langle f, x \rangle$$

where f is an element of the dual space of the Lie algebra, i.e. a real linear functional on the algebra, and $\langle f, x \rangle$ denotes the evaluation of f at x .

Remark. Note that we omit the 2π to maintain clarity of the following calculations.

Theorem 5.4.1. *A map on the Lie algebra $x \mapsto i\langle f, x \rangle$, defines a one-dimensional representation if and only if*

$$(f, [x, y]) = 0$$

Proof. The proof of necessity is due to Kirillov [7, pp. 72]. Suppose dU is a one dimensional representation of a Lie algebra with $dU : x \mapsto i\langle f, x \rangle$. Then $dU([x, y]) = [dU(x), dU(y)] = [\lambda_x, \lambda_y] = 0$. So $dU([x, y]) = i\langle f, [x, y] \rangle = 0$. For the converse assume $i\langle f, [x, y] \rangle = 0$. $x \mapsto i\langle f, x \rangle$ is a Lie algebra representation if $[x, y] \mapsto [i\langle f, x \rangle, i\langle f, y \rangle] = i\langle f, [x, y] \rangle$. Notice then

$$[i\langle f, x \rangle, i\langle f, y \rangle] = i\langle f, x \rangle i\langle f, y \rangle - i\langle f, y \rangle i\langle f, x \rangle$$

as $\langle f, x \rangle$ is simply a value in the field over which the representation is defined. $i\langle f, x \rangle i\langle f, y \rangle - i\langle f, y \rangle i\langle f, x \rangle = 0$ by field axioms. Therefore we conclude that

$$[i\langle f, x \rangle, i\langle f, y \rangle] = i\langle f, [x, y] \rangle$$

so $dU : x \mapsto i\langle f, x \rangle$ is a one-dimensional representation $\iff (f, [x, y]) = 0$. $\square$

So we may conclude that all one-dimensional representations of the Lie algebra are of the form $x \mapsto i\langle f, x \rangle$ where $(f, [x, y]) = 0$.

It is natural then that given $f \in \mathfrak{g}^*$ we might be interested in subalgebras $\mathfrak{g}_1$ of $\mathfrak{g}$ where for all $x, y \in \mathfrak{g}_1$, we have $(f, [x, y]) = 0$, as we know by theorem 5.4.1 that $x \mapsto i\langle f, x \rangle$ will then define one-dimensional representation of the subalgebra $\mathfrak{g}_1$ and hence a one-dimensional representation of $G_1 = \exp(\mathfrak{g}_1)$ given by

$$U_{\exp(x)} = e^{i\langle f, x \rangle}.$$

Given this situation we say $\mathfrak{g}_1$ is *subordinate* to f , in fact the conditions

$$\bullet F|_{[\mathfrak{b}, \mathfrak{b}]} = 0$$

- $X \mapsto i\langle F, X \rangle$ defines a character of $\mathfrak{h}$.

are equivalent.[7, pp. 72] Further, when

$$\text{codim}_{\mathfrak{g}} \mathfrak{h} = \frac{1}{2} \text{rk} B_F$$

(where B_F is the antisymmetric bilinear form $B_F(X, Y) = \langle F, [X, Y] \rangle$) we call $\mathfrak{h}$ a real algebraic polarisation. Real algebraic polarisations always exist for nilpotent Lie algebras.

5.5 Coadjoint orbits for Heisenberg group

Here we present the construction of the irreducible unitary representations of the three dimensional Heisenberg group using the orbit method.

The Heisenberg group H is defined as:

$$H := \left\{ \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}, (a, b, c) \in \mathbb{R}^3 \right\} \subset GL_3(\mathbb{R})$$

The Lie algebra, denoted $\mathfrak{h} = \text{Lie}(H)$, is comprised of matrices of the form:

$$\mathfrak{h} := \left\{ \begin{pmatrix} 0 & a & c \\ 0 & 0 & b \\ 0 & 0 & 0 \end{pmatrix}, (a, b, c) \in \mathbb{R}^3 \right\} \subset GL_3.$$

The matrices

$$X = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

form a basis for the Lie algebra $\mathfrak{h}$. The commutation relation is $[X, Y] = XY - YX = Z$.

By theorem 4.1.2

$$\mathfrak{h}^* \cong \text{Mat}_3(\mathbb{R})/\mathfrak{h}^\perp.$$

Recall

$$\mathfrak{h}^\perp = \{A \in \text{Mat}_3(\mathbb{R}) \mid (A, B) = 0, \quad \forall B \in \mathfrak{h}\}$$

$$\mathfrak{h}^\perp \cong \left\{ \begin{pmatrix} x_1 & x_2 & x_3 \\ 0 & x_4 & x_5 \\ 0 & 0 & x_6 \end{pmatrix}, (x_i) \in \mathbb{R}^3 \right\}$$

then,

$$\mathfrak{h}^* \cong \text{Mat}_3(\mathbb{R})/\mathfrak{h}^\perp \cong \left\{ \begin{pmatrix} 0 & 0 & 0 \\ x & 0 & 0 \\ z & y & 0 \end{pmatrix}, (x, y, z) \in \mathbb{R}^3 \right\}.$$

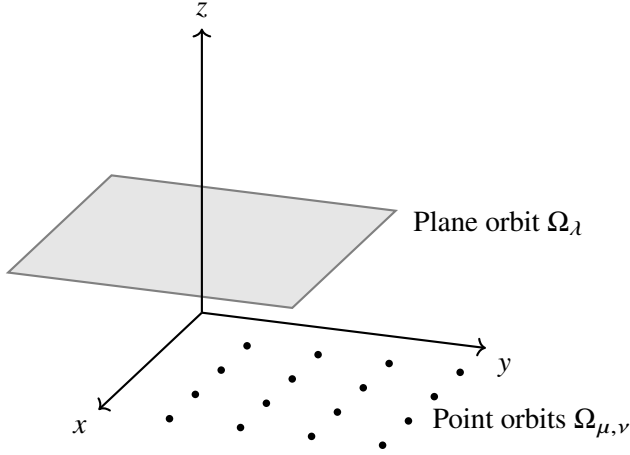


Figure 5.1: Heisenberg group coadjoint orbits

Given $\beta \in \mathfrak{h}^*$ and $g \in H$,

$$K(g)\beta = p_V(g\beta g^{-1})$$

as matrices:

$$\begin{aligned} & \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ x & 0 & 0 \\ y & z & 0 \end{pmatrix} \begin{pmatrix} 1 & -a & ab-c \\ 0 & 1 & -b \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} ax + cz & cy - a(ax + cz) & (ab - c)(ax + cz) - bcy \\ bz + x & by - a(bz + x) & (ab - c)(bz + x) - b^2y \\ z & y - az & z(ab - c) - by \end{pmatrix}. \end{aligned}$$

The projection for the Heisenberg algebra amounts to sending all non-lower triangular elements to 0

$$K(g)\beta = \begin{pmatrix} 0 & 0 & 0 \\ bz + x & 0 & 0 \\ z & y - az & 0 \end{pmatrix}.$$

The coadjoint orbit corresponding to β is given by

$$\mathcal{O}(\beta) = \{K(g)\beta | g \in H\}.$$

Recalling that all entries are arbitrary real numbers, we see that there are two kinds of orbits, those where $z = 0$ and where $z \neq 0$. See when $z = 0$ the coadjoint action is trivial, so the orbit is simply a point parameterised by x, y . When $z \neq 0$, the coadjoint action defines a plane intersecting the point z . The plane orbits Ω_λ are isomorphic to $\mathbb{R}^2$.

5.6 Calculation of canonical symplectic form on the orbit

Plane orbit 2-form

The canonical differential 2-form on the orbit Ω_λ is given by

$$\sigma_\Omega(\beta)(K_*(X)\beta, K_*(Y)\beta) = \langle \beta, [X, Y] \rangle$$

where K_* is the coadjoint representation of the Lie algebra.

The image of the basis vectors for the Lie algebra X, Y, Z under the coadjoint action are given by:

$$K_*(X) = -z\partial_y \quad K_*(Y) = z\partial_x \quad K_*(Z) = 0.$$

The symplectic form satisfies:

$$\sigma_\Omega(\beta)(K_*(X)\beta, K_*(Y)\beta) = \langle \beta, [X, Y] \rangle = \langle \beta, Z \rangle = z.$$

Writing the symplectic form in the canonical local coordinates, $\sigma_\Omega = f_1 dx \wedge dy$, we see that $\sigma(\beta)(K_*(X)\beta, K_*(Y)\beta) = f_1 dx \wedge dy = z$ so we can solve for f_1 .

$$f_1 = \frac{1}{z}$$

and so

$$\sigma(\beta) = \frac{1}{z} dx \wedge dy. \tag{5.3}$$

Point orbit 2-form

We can also calculate the canonical 2-form for the point orbits $\omega_{\mu, \nu}$. Given a functional

$\beta = \begin{pmatrix} 0 & 0 & 0 \\ \mu & 0 & 0 \\ 0 & \nu & 0 \end{pmatrix}$, the orbit is a point orbit $\Omega_{(\mu, \nu)}$, the infinitesimal generators will return 0,

$$K_*(X) = 0 \quad K_*(Y) = 0 \quad K_*(Z) = 0.$$

The symplectic form is trivial as the point can be considered as $\mathbb{R}^0$,

$$\sigma_\Omega(\beta)(K_*(X)\beta, K_*(Y)\beta) = 0.$$

The volume form associated with $\omega_{\mu, \nu}$ is the 0-form $\omega_{\mu, \nu}^0$, we know by definition that a volume form is never 0, causing us to set $\omega_{\mu, \nu}^0 = 1$ (the constant function $f(x) = 1$). The measure μ_ω associated with $\omega_{\mu, \nu}$ will then be

$$\mu_\omega(\xi) = \int_\xi \omega_{\mu, \nu} = \int_\xi 1.$$

The integral of a 0-form is simply evaluation at the point over which we are integrating, so

$$\mu_\omega(\xi) = \int_\xi \omega_{\mu, \nu} = \int_\xi 1 = 1.$$

So we conclude that the measure is the dirac delta centred at the point ξ .

5.7 Construction of unitary irreducible representations

The Heisenberg group is a nilpotent group with one-dimensional centre, so the theory applies directly. We conclude that all the unitary irreducible representations are induced from a one-dimensional representation $e^{i\langle 2\pi i\beta, x \rangle}$ of a subgroup subordinate to the linear functional β . The action of β on $X \in \mathfrak{g}$ is given by $\langle \beta, X \rangle = \text{tr}(\beta X)$.

We begin with the 2-dimensional orbit Ω_λ . Choose $F \in \Omega_\lambda$. We must find a subalgebra $\mathfrak{a} \subset \mathfrak{h}$ such that $F|_{[\mathfrak{a}, \mathfrak{a}]} = 0$. Notice that $\mathfrak{a} = yY \oplus zZ$ is abelian and so $[\mathfrak{a}, \mathfrak{a}] = 0$, therefore $\mathfrak{a}$ is subordinate to every linear functional F . Further calculating

$$\text{codim}_{\mathfrak{h}} \mathfrak{a} = 1$$

and

$$\text{rk} B_F = \dim \mathfrak{h} - \dim(\text{stab}(F)) = 3 - 1 = 2$$

so $\text{codim}_{\mathfrak{h}} \mathfrak{a} = \frac{1}{2} \text{rk} B_F$, confirming the subalgebra is of maximal dimension.

Next we define $A = \exp \mathfrak{a}$, and write the one-dimensional representation of A as

$$\rho_{F,A}(\exp X) = e^{2\pi i \langle F, X \rangle}$$

Choosing $F \in \Omega_\lambda$ to be

$$\begin{pmatrix} 0 & 0 & 0 \\ x' & 0 & 0 \\ \lambda & 0 & 0 \end{pmatrix},$$

then for $X \in \mathfrak{a}$,

$$X = \begin{pmatrix} 0 & 0 & \gamma \\ 0 & 0 & \beta \\ 0 & 0 & 0 \end{pmatrix}.$$

Then $\langle F, X \rangle = \lambda\gamma$ chosen to ensure the functional is in a plane orbit intersecting the Z axis at λ , and the representation

$$\rho_{F,A}(\exp X) = e^{2\pi i \lambda \gamma}.$$

We then follow the induction procedure outlined in section 3.4 and [3, pp. 66].

As A is a Lie subgroup of H , we may form the homogeneous manifold $M = H/A$. As above we associate H with $\mathbb{R}^3$ and A with $\mathbb{R}^2$ such that $M \leftrightarrow \mathbb{R}$. The sections

$$s : M \rightarrow H,$$

are given by

$$x \mapsto \exp(xX).$$

The master equation is written

$$s(x)g = h(x, g)s(x \cdot g).$$

We calculate the $h(x, g)$ term:

$$h(x, g) = \begin{pmatrix} 1 & x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -y & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & a+x-y & c+xb \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix},$$

which can be written:

$$s(x) \cdot g = \exp(a) s(x \cdot g).$$

We know however by definition that $h(x, g) \in A$, therefore we conclude that $x \cdot g = a + x$. In matrix form the master equation reads

$$\begin{pmatrix} 1 & x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & c+xb \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & y & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

So we find the infinite dimensional representation of the whole group H on the space $L^2(H/A, \mu)$.

Let

$$\exp \kappa = \begin{pmatrix} 1 & 0 & c+xb \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix},$$

then

$$((\text{Ind}_A^H \rho)(g)f)(x) = e^{2\pi i \langle F, \kappa \rangle} f(x \cdot g) = e^{2\pi i \lambda (c+bx)} f(x+a) \quad (5.4)$$

as $y := x \cdot g$ in the master equation. The irreducibility of this representation is then guaranteed by theorem 5.1.2.

5.8 Generalised characters of the Heisenberg group

We know the unitary dual $\hat{H}$ breaks into one-dimensional $\pi_{\mu, \nu}$ and infinite dimensional π_λ parts, the character theory is then only non-trivial for the infinite dimensional characters π_λ corresponding to plane orbits through $z = \lambda$. The characters are given by

$$\text{tr } \pi_\lambda(\phi) = \int_{\mathbb{R}^2} \tilde{\phi}(x, y, \lambda) \frac{dx \wedge dy}{\lambda}$$

the derivation can be found in [3, §3.2].

When we associate the plane orbit Ω_λ through $z = \lambda$ with $\mathbb{R}^2$ and substitute the KKS form $\omega = \frac{dx \wedge dy}{\lambda}$ the Kirillov character formula

$$\chi = \int_{\Omega} \tilde{\phi}(f) d\omega(f)$$

agrees with the previous result.

5.9 Geometric description of induction and restriction

We can describe the induction and restriction of representations via the projection of the coadjoint orbit onto a subalgebra in $\mathfrak{g}^*$.

The subalgebras of the Heisenberg group are:

- Lines $\mathbb{R} \cdot X$
- Planes containing the centre $xX \oplus zZ$
- The whole algebra $\mathfrak{h}$.

Example 5.9.1. Given $F_{x,y,z}$ with $z = 0$ consider the one dimensional representation of H given by

$$\pi_{\mu,\nu} = e^{2\pi i \langle F, X \rangle}.$$

Consider the restriction of this representation to the subalgebra $A = xX \oplus zZ$, written $\text{Res}_A^H \rho(\exp X)$. According to the orbit method we write

$$\text{Res}_A^H \pi_{\mu,\nu} = \int_{\omega \in p((\mu,\nu))} m(\Omega, \omega) \pi_\omega d\omega.$$

The projection of the point orbit (μ, ν) onto $\mathfrak{a}^*$ will simply be the point μ on the x axis. Write $\xi = (\mu, 0, 0)$ to denote the point in $\mathfrak{h}^* \leftrightarrow \mathbb{R}^3$.

$$\int_{\xi} m(\Omega, \omega) \pi_{(\mu,0)} d(\mu, 0) = \int_{\xi} m(\Omega, \omega) e^{2\pi i \langle \mu, 0 \rangle} d(\mu, 0).$$

The graph Γ_p will be the plane $xX \oplus zZ$, calculating

$$k((\mu, \nu), (\mu, 0)) = \dim(xX \oplus zZ \cap (\mu, \nu), (\mu, 0)) = 0.$$

So we conclude $m((\mu, \nu), (\mu, 0)) = 1$ and write

$$\int_{\xi} 1 \cdot e^{2\pi i \langle (\mu,0), X \rangle} d(\mu, 0).$$

As above, the measure $d(\mu, 0)$ is the dirac delta $\delta_{\mu,0}$. Then using the properties of the dirac delta we calculate

$$\int_{\xi} 1 \cdot e^{2\pi i \langle (\mu,0), X \rangle} d(\mu, 0) = e^{2\pi i a \mu}.$$

This is the representation $\pi_{\mu,0}$.

This agrees with the regular definition of the restriction,

$$\text{Res}_A^H e^{2\pi i a \mu + b \nu} = e^{2\pi i a \mu + b \nu}|_A.$$

Recalling that elements of the group are written $h = aX + bY + cZ$ to see that

$$e^{2\pi i a \mu + b \nu}|_A = e^{2\pi i a \mu}.$$

Chapter 6

Compact Groups

Remarkably, despite its origins in nilpotent groups, the ideology of the orbit method (that unitary irreducible representations are in correspondence with coadjoint orbits) can be inhabited to explain the representation theory of compact groups. The orbit method rephrases the extant Cartan-Weyl theory in geometric language, enabling new insight if not new theorems.

6.1 Preliminary definitions

Maximal Torus

Definition 6.1.1. The maximal torus of a compact connected Lie group G is an abelian Lie subgroup T of G such that T is maximal, that is any other Lie abelian Lie subgroup is contained in or equal to T .

Theorem 6.1.1. *Any element g of a compact connected Lie group G is conjugate to the maximal torus T .*

Weyl group

Definition 6.1.2. [18, Definition 7.25]

The Weyl group is the group of symmetries of the root system ϕ . In more detail, to each root $\alpha \in \phi$ define

$$s_\alpha(v) = v - 2 \frac{\langle \alpha, v \rangle}{\langle \alpha, \alpha \rangle} \alpha$$

The group generated by the s_α transformations with $\alpha \in \phi$ is called the *Weyl group*.

The s_α transformations are reflections around hyperplanes orthogonal to the roots. Recall that the roots of a compact Lie algebra $\mathfrak{h}$ can be understood as linear functionals in $\mathfrak{h}^*$. We can associate a functional $F \in \mathfrak{h}^*$ with an element $X_F \in \mathfrak{h}$ via the formula

$\langle X_F, Y \rangle = (X, Y)_K$ for all $Y \in \mathfrak{h}$. In the same way we can then associate the space $i\mathfrak{t}^*$ spanned by roots with a subspace of $i\mathfrak{t}[3, \S 5.2.1]$.

Definition 6.1.3. A Weyl chamber is a connected component of the complement of the Weyl set of hyperplanes.

Definition 6.1.4. The positive Weyl chamber is defined to be the set

$$C_+ = \{\lambda \in \mathbb{R} \mid (\lambda, \alpha) \geq 0 \quad \forall \alpha \in \phi_+\}.$$

All weights in the positive Weyl chamber are dominant weights.

Integral elements

Given a semisimple Lie algebra $\mathfrak{g}$ with Cartan subalgebra $\mathfrak{h}$, a functional $\lambda \in \mathfrak{h}^*$ may be exponentiated to a function on $H = \exp \mathfrak{h}$ written

$$e^\lambda(\exp X) := e^{\lambda(X)}.$$

This function is not necessarily well defined as the exponential map is not always injective. Suppose that we have $X_1 \neq X_2 \in \mathfrak{h}$, with $\exp X_1 = \exp X_2$. Evaluating e^λ we see

$$e^{\lambda(X_1)} = e^\lambda(\exp X_1) \neq e^\lambda(\exp X_2) = e^{\lambda(X_2)}$$

unless $e^{\lambda(X_1)} = e^{\lambda(X_2)}$.

Definition 6.1.5. Given the kernel of the exponential map

$$\Gamma = \{X \in \mathfrak{h} \mid \exp X = 1 \in H\},$$

we call an element λ analytically integral if $\lambda(X) \in 2\pi i\mathbb{Z}$ for all $X \in \Gamma$.

We offer some motivation for this definition: Given X_1, X_2 in $\mathfrak{h}$ with $X_1 \neq X_2$ and $\exp X_1 = \exp X_2$, in order for e^λ to be well defined we must have

$$e^\lambda(\exp X_1) = e^\lambda(\exp X_2)$$

or equivalently

$$e^{\lambda(X_1)} = e^{\lambda(X_2)}.$$

However $X_1 \neq X_2$ so this is not immediately true. In order for the relation to hold, we must have

$$\lambda(X_1) = \lambda(X_2) \pm 2\pi i\mathbb{Z}$$

which by linearity of λ implies

$$\lambda(X_1 - X_2) \in 2\pi i\mathbb{Z}.$$

Notice that as $\mathfrak{h}$ is abelian the exponential map is a homomorphism,

$$\exp(X_1 - X_2) = \exp(X_1) \cdot \exp(-X_2)$$

by a standard property of the exponential map $\exp(-X_2) = \exp(X_2)^{-1}$. Therefore,

$$\exp(X_1) \cdot \exp(-X_2) = \exp(X_1) \cdot \exp(X_2)^{-1}$$

using the equality $\exp(X_1) = \exp(X_2)$ we conclude

$$\exp(X_1) \cdot \exp(X_2)^{-1} = 1,$$

therefore $(X_1 - X_2) \in \Gamma$. From these calculations we see that the function e^λ is only well defined for functionals λ satisfying the analytic integrality condition. The analytically integral (hereafter integral) weights are the weights which define a character of the stabiliser subgroup. There is another notion of integrality called *algebraic integrality* for simply connected compact groups the sets of algebraically integral elements and analytically integral elements coincide. The integral weights form a lattice denoted $P(G)$ which forms the set of all weights of all finite dimensional holomorphic representations of G . The weight lattice is freely generated by the fundamental weights ω_j , which are defined by the relation $\frac{2(\omega_j, \alpha_i)}{(\alpha_i, \alpha_i)} = \delta_{ij}$.

6.2 Coadjoint orbits

We begin with a lemma from [3, Chapter 5, Lemma 3]:

Lemma 6.2.1. *Let C_+ be the positive Weyl chamber in $\mathfrak{it}^*$. Any K -orbit in $\mathfrak{k}^* \cong \mathfrak{k}$ intersects the set iC_+ in exactly one point.*

This lemma allows us to label the coadjoint orbits with elements of the positive Weyl chamber C_+ . Consider the following corollary of the highest weight theorem,

Corollary 6.2.1.1. *For a compact simply connected group K the set $\hat{K}$ of all unitary irreducible representations is labelled by the set of dominant integral weights.*

The obstruction lies in the identities

$$\begin{aligned} C_+ &\cong \mathbb{R}_+^n \\ \text{Set of dominant integral weights} &\cong \mathbb{Z}_+^n. \end{aligned}$$

This leads us to require a condition on the orbits to cut down the admissible orbits, we call this condition the integrality condition.

6.3 Integrality condition

The coadjoint orbits correspond to the elements of a positive Weyl chamber, and the integral elements of a positive Weyl chamber correspond to irreducible representations. So given a coadjoint orbit, we would like to find some condition intrinsic to the orbit that determines if the orbit is associated to an integral element λ and hence defines a character of a subgroup. This condition is the integrality condition. In accordance with the ideology of the orbit method, the integrality condition admits a geometric formulation.

Theorem 6.3.1 (Kirillov [3]). *The following are equivalent:*

- *The orbit $\Omega \subset \mathfrak{g}^*$ is integral*
- *$\int_C \sigma \in \mathbb{Z}$ for every geometric 2-cycle C in Ω ¹*
- *There exists a line bundle over Ω with a Hermitian connection ∇ such that $\text{curv}(\nabla) = 2\pi i\sigma$*
- *For any $F \in \Omega$ the 1-dimensional representation of the Lie algebra $\text{stab}(F)$ given by $X \mapsto 2\pi i\langle F, X \rangle$ integrates to a character of the connected component of the identity in $\text{Stab}(F)$*

In the programme of geometric quantisation where we consider a symplectic manifold (M, ω) , the integrality of ω corresponds to the manifold M being 'quantisable', see Hall [30, Chapter 25].

Corollary 6.3.1.1. *Every integral orbit passes through an integral weight in the maximal torus.*

Proof. We can deduce corollary 6.3.1.1 given two facts from [3, Chapter 5 §2.1]:

- *The stabiliser of any point $X \in \mathfrak{k} \cong \mathfrak{k}^*$ is conjugate to a subgroup S which contains the maximal torus T .*
- *For any point F in a (co)adjoint orbit of maximal dimension, the stabiliser is conjugate to the maximal torus T*

From these facts along with the integrality condition we can conclude that for an integral element λ , the representation

$$\chi(\exp X) = e^{2\pi i\langle \lambda, X \rangle}$$

is a representation of the maximal torus T or a representation of a group $S \supset T$ which can be restricted to a rep of T . This representation is well defined if and only if λ is an analytically integral element.

□

¹ See the appendix for a brief introduction to cycles.

6.4 Proof of correspondence

The proof of the orbit method then lies in showing that the integral orbits (given by the integrality condition below) are also labelled by the dominant integral weights P_+ and hence correspond to the irreducible representations of the group.

Theorem 6.4.1. *For a compact, simply connected Lie group K , the coadjoint orbits lie in one-to-one correspondence with the irreducible representations.*

Proof. Recalling theorem 6.2.1 we see that the set of coadjoint orbits is labelled by points in $\mathbb{R}^n$. These points, by definition of the positive Weyl chamber are all dominant weights. From the highest weight theorem we know that the set of irreducible representations is indexed by the dominant integral weights. By corollary 6.3.1.1 above, we see the set of integral orbits written O_{int} all pass through integral weights. Thus we can conclude that any integral orbit passes through $\mathfrak{t}$ at a dominant integral weight, establishing the correspondence between integral coadjoint orbits and integral dominant weights and hence the irreducible representations of the compact group K . $\square$

Labelling problem

To denote a coadjoint orbit passing through the weight λ , we write Ω_λ . To this orbit we associate the representation

$$\rho_{\lambda,H}(\exp X) = e^{2\pi i \langle \lambda, X \rangle}$$

of the subgroup $H = \text{Stab}(\lambda)$, thus establishing the pairing

$$\Omega_\lambda \leftrightarrow \rho_{\lambda,H}.$$

This notation causes some discord between the orbit method and the customary notation used in other literature. This is well expressed by the example of $SU(2)$. Any representation π_m of $SU(2)$ has $m + 1$ weights with highest weight m , where a weight $m \in \mathfrak{t}^*$ satisfies

$$\pi_m(X) \cdot v = m(X) \cdot v.$$

We can diagonalise the representation π_m on the torus T according to the weights and using the standard parameterisation of elements in the the torus $T_\theta = (e^{i\theta}, e^{-i\theta})$,

$$\pi_m(T) = \begin{pmatrix} e^{im\theta} & 0 & \dots & 0 \\ 0 & e^{i(m-2)\theta} & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & e^{-im\theta} \end{pmatrix}$$

This is the standard notation for the weights of $SU(2)$, where we label the representation π_m according to the highest weight m .

Taking the orbit Ω_m , the correspondence should read:

$$\Omega_m \leftrightarrow e^{im} \leftrightarrow \pi_m$$

however at present the orbit Ω_m will be associated with the representation $\rho_{m,T}(\exp X) = e^{2\pi i m(X)}$. To remedy this we make the notational correction writing

$$\Omega_\lambda \leftrightarrow \text{orbit through } \frac{\lambda}{2\pi i}$$

so that the representation associated with the orbit Ω_m is $\rho_{m,T}(\exp X) = e^{im(X)}$.

6.5 Character formula

The character formula of Kirillov is an equivalent formulation of the correspondence developed above. In the case of compact groups the Kirillov formula follows from the Weyl character formula [3, Chapter 5, theorem 1]:

$$\chi_\lambda(t) = \frac{\sum_{w \in W} (-1)^{\ell(w)} e^{w(\lambda+\rho)}(t)}{\sum_{w \in W} (-1)^{\ell(w)} e^{w(\rho)}(t)} \quad t \in T, \quad (6.1)$$

and a result from Harish-Chandra [24], [11]. In [24] it is shown that the Kirillov character formula

$$\chi_\lambda(\exp X) = \frac{1}{j(X)} \int_{\Omega_\lambda} e^{2\pi i \langle F, X \rangle} \sigma(F),$$

is independent of the Weyl formula, where χ_λ is the character of the irreducible representation π_λ with highest weight λ and σ is the canonical 2-form on Ω . The right hand side can be understood as the Fourier transform of the measure σ over the orbit that passes through $\frac{\lambda}{2\pi i}$ where we use the correspondence $\Omega_\lambda \leftrightarrow \frac{\lambda}{2\pi i}$.

The $\frac{1}{j(X)}$ term

In flat Euclidean space, the Lebesgue measure is a Haar measure. The Lie algebra $\mathfrak{g}$ lives in Euclidean space, and hence admits the Lebesgue measure as a Haar measure. For unimodular groups when we exponentiate the measure on the Lie algebra to find a measure on the group, we pick up a factor $p(X)$, the Jacobian of the transformation. The Fourier transform of a function $f(X)$ on the Lie algebra $\mathfrak{g}$ is given by

$$\hat{\phi}(f) = \int_{\mathfrak{g}} \phi(X) e^{2\pi i \langle f, X \rangle} dX.$$

We use this definition to define a Fourier transform of a function $\varphi(g)$ on the group G ,

$$\hat{\varphi}(f) = \int_G \varphi(\exp X) e^{2\pi i \langle f, X \rangle} dg,$$

where these definitions are given in terms of distributions, but coincide with regular Fourier transforms in the usual way [7, pp. ~ 86–89]. For a general Lie group, the measures

dX and dg are related by the factor $p(X)$, for compact groups this factor is exactly

$$p(X) = \det \frac{\sin(\operatorname{ad}(X/2))}{\operatorname{ad}(X/2)} \quad (6.2)$$

Then $j(X) := p(X)^{\frac{1}{2}}$.

6.6 Construction of unitary irreducible representations

The construction of the representations corresponding to integral coadjoint orbits follows the programme of geometric quantisation pioneered by Kostant and Souriau. We discuss the theory of geometric quantisation in the next chapter and compute the explicit example of $SU(2)$ in the final chapter.

Chapter 7

Geometric Quantisation

"The procedure (of geometric quantisation) is admittedly somewhat magical"

N.J. Wildberger

We now review the necessary framework of Hamiltonian mechanics and geometric quantisation as developed by Kostant and Souriau. Much of this material has been found in [30], [13], [28] and [12] and has been cited explicitly where definitions or theorems have been lifted verbatim with the exception of the correction of typographical inaccuracies.

Geometric quantisation is a method for quantising a classical mechanical system, i.e. finding an irreducible representation of the algebra of functions on a symplectic manifold that respects the Poisson bracket. When the symplectic manifold admits a G action (where G is a Lie group) that preserves the symplectic structure, we refer to the manifold as a *Hamiltonian G -manifold*. Coadjoint orbits are very natural Hamiltonian G -manifolds, warranting their study via geometric quantisation.

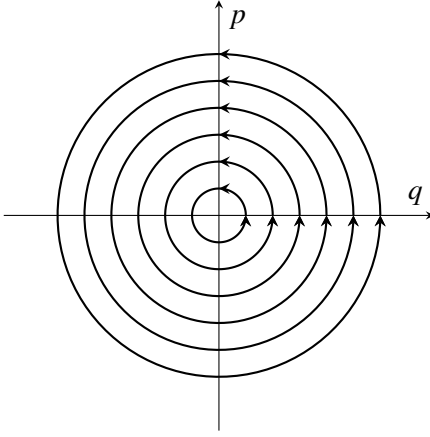
The procedure can be broken into two parts, *pre-quantisation* and *quantisation*. Pre-quantisation establishes a representation on a Hilbert space associated with the symplectic manifold and quantisation uses a polarisation to make this representation irreducible.

7.1 Prequantisation

Let M be a symplectic manifold and H a Hilbert space. The pre-quantisation procedure aims to develop a homomorphism

$$Q_{pre} : C^\infty(M) \rightarrow \mathcal{B}(H). \quad (7.1)$$

This is a representation of the Poisson algebra of functions on the manifold. The most basic setting for geometric quantisation is when the manifold is a cotangent bundle T^*M with

Figure 7.1: Flow ϕ_t of the Harmonic Oscillator X_H

natural projection

$$\pi : T^*M \rightarrow M.$$

Cotangent bundles arise naturally in the consideration of phase spaces and are key examples of symplectic manifolds. We introduce some terminology through a simple example.

Example 7.1.1. [28, pp. 142] Suppose we have a phase space describing the position and momentum of a particle, we will use $M = \mathbb{R}^2$ as our example with points $x \in M$ described by their position and momentum $x = (q, p)$, we will see later that a phase space need not be a flat Euclidean space. The Hamiltonian is a function

$$H : M \rightarrow \mathbb{R}$$

that represents the total energy of a point. The canonical two-form on the cotangent space is $\omega = dp \wedge dq$. The Hamiltonian vector field is then given by

$$X_H = \left(\frac{\partial H}{\partial p}, -\frac{\partial H}{\partial q} \right),$$

the flow lines of which represent states with constant energy, i.e. energy is not being added or removed.

Functions $f \in C^\infty(M)$ are called observables, for example

$$f(p, q) = p$$

is the momentum observable. These functions form a Lie algebra under the Poisson bracket

$$\{f, g\} = \frac{\partial f}{\partial q} \frac{\partial g}{\partial p} - \frac{\partial f}{\partial p} \frac{\partial g}{\partial q}.$$

An observable $f \in C^\infty(M)$ is called a constant of motion if

$$\{f, H\} = 0,$$

in this way the symmetries of the system are revealed.

Suppose

$$H(p, q) = \frac{1}{2} (p^2 + k^2 q^2),$$

then

$$X_H = \left(\frac{\partial H}{\partial p}, -\frac{\partial H}{\partial q} \right) = (p, -k^2 q).$$

Consider the total energy observable

$$H(q, p)$$

then

$$\{H, H\} = 0.$$

So H is constant along the flow lines of the Hamiltonian vector field i.e. any constant of motion is invariant under the action of some group for which the orbit is the flow of the vector field.

Further consider the position and momentum observables

$$g(q, p) = q \quad f(q, p) = p,$$

then

$$\{g, H\} = \frac{\partial g}{\partial q} p - \frac{\partial g}{\partial p} k^2 q = p,$$

and

$$\{f, H\} = \frac{\partial f}{\partial q} p - \frac{\partial f}{\partial p} k^2 q = -k^2 q.$$

So one can see that the flow lines are given by orbits of the action of $U(1)$ on the manifold M .

More generally,

Definition 7.1.1. Suppose we have a symplectic manifold M , ω , and the space of functions $C^\infty(M)$, the Poisson bracket on $C^\infty(M)$ is given by

$$\{f, g\} := \omega(X_f, X_g)$$

where X_f is the Hamiltonian vector field associated with the function f that is, X_f is the vector field that satisfies

$$df(Y) = \omega(X_f, Y)$$

for all vector fields Y on M .

Lemma 7.1.1. The functions $f \in C^\infty(M)$ form a Lie algebra with the Poisson bracket as the commutator.

Pre-quantisation map

Dirac posed three requirements of a quantisation map Q [13, §8.1]:

- Q should be $\mathbb{R}$ linear
- If f is constant then $Q(f)$ should be a multiplication operator
- Q should preserve the Poisson bracket, $iQ(\{f, g\}) = [Q(f), Q(g)]$.

The quantum mechanical counterpart of an observable is a self adjoint operator. The factor of i is introduced in requirement 3 as given two self adjoint operators $Q(f), Q(g)$, $[Q(f), Q(g)]^* = -[Q(f), Q(g)]$ the factor of $\frac{1}{i}$ rectifies this. One can define the Lie bracket to include the $\frac{1}{i}$ term, the $Q(f)$ operators remain a Lie algebra under this new bracket.

Remark. Note that we omit the 2π and set $\hbar = 1$ to maintain agreement with the majority of the literature. The inclusion of these factors will scale the curvature of the covariant derivative by $2\pi\hbar$.

Cotangent manifolds

A natural point of departure in the attempt to find a map Q_{pre} is the space of sections of the trivial line bundle

$$L = M \times \mathbb{C},$$

the space of square integrable sections of L is the Hilbert space $L^2(\mathbb{R}^{2n})$. Given some $f \in C^\infty(M)$, the vector field X_f defined as in definition 7.1.1 is a differential operator on $L^2(\mathbb{R}^{2n})$. A natural first choice for the quantisation map is then

$$f \mapsto X_f,$$

this however fails requirement 2 as for f constant, $X_f = 0$. Correcting this failure leads us to write the pre-quantisation map as:

$$Q_{pre}(f) = i(X_f - i\theta(X_f)) + f$$

where θ is the canonical one-form on the cotangent space given by

$$d\theta = \omega.$$

The $X_f - i\theta(X_f)$ term in the quantisation map has some interesting properties. Write the covariant derivative as

$$\nabla_X = X - i\theta(X)$$

where X is a vector field on T^*M and calculate

$$[\nabla_{X_f}, \nabla_{X_g}] = \nabla_{X_{\{f, g\}}} + i\{f, g\} = \nabla_{X_{\{f, g\}}} + i\omega(X_f, X_g).$$

The two-form $i\omega$ is called the *curvature* of the covariant derivative ∇_X . The quantisation map then reads

$$Q_{pre}(f) = i\nabla_{X_f} + f$$

and

$$Q_{pre}(\{f, g\}) = i\nabla_{X_{\{f, g\}}} + \{f, g\}.$$

Lemma 7.1.2 ([30, Proposition 22.1]). . For all $f, g \in C^\infty(\mathbb{R}^{2n})$,

$$[Q_{pre}(f), Q_{pre}(g)] = iQ_{pre}(\{f, g\})$$

It is exactly the curvature term $i\omega$ which ensures that the above holds.

Example 7.1.2. Returning to example 7.1.1, let $f(p, q) = p$ be the momentum observable with vector field

$$X_f = \frac{\partial f}{\partial q} \frac{\partial}{\partial p} - \frac{\partial f}{\partial p} \frac{\partial}{\partial q} = -\frac{\partial}{\partial q}$$

and write canonical one form θ as

$$\theta(X_f) = (pdq)X_f.$$

The prequantisation map is

$$Q_{pre}(f) = i \left(-\frac{\partial}{\partial q} - i(pdq) \frac{\partial}{\partial q} \right) + f = -i \frac{\partial}{\partial q} + p + f.$$

Let $g \in L^2(\mathbb{R}^2)$, then

$$Q_{pre}(f)g(p, q) = -i \frac{\partial}{\partial q} g(p, q) + pg(p, q) + fg(p, q) = -i \frac{\partial}{\partial q} g(p, q).$$

which is exactly the one-dimensional quantum mechanical momentum operator.

General symplectic manifolds

We now move to describe these constructions on non cotangent bundle. As above given the symplectic manifold M we form the complex line bundle L over M

$$\pi : L \rightarrow M,$$

which admits a local trivialisation $U \subset M$ such that there is a diffeomorphism $\chi : \pi^{-1}(U) \rightarrow M \times \mathbb{C}$. This is the statement that locally line bundle looks like $M \times \mathbb{C}$. The space of sections on the line bundle consists of smooth maps

$$s : M \rightarrow L,$$

such that $\pi(s(p)) = p$ for $p \in M$. One can see that this construction is almost identical to the cotangent bundle case, however the line bundle is not trivial. As such the construction of the canonical symplectic one-form θ fails in this case and we must develop a suitable corresponding notion for the non-trivial line bundle.

Definition 7.1.2. [30, Definition 23.2] A connection ∇ on a line bundle L over M is a map associating to each vector field X on M and a section s of L to another section $\nabla_X(s)$ of L satisfying:

- $\nabla_{fX}(s) = f\nabla_X(s)$
- $\nabla_X(fs) = X(f)(s) + f\nabla_X(s)$

for $f \in C^\infty(M)$.

After choosing a vector field X , the operator ∇_X is referred to as the covariant derivative.

Definition 7.1.3. [30, Definition 23.3] A Hermitian structure on a line bundle L over M is a choice of an inner product $(\cdot, \cdot)$ on each fibre $\pi^{-1}(\{x\})$ of L such that for each smooth section s of L , (s, s) is a smooth function on M . A line bundle L together with a choice of Hermitian structure on L is then called a Hermitian line bundle. A connection ∇ on a Hermitian line bundle is called Hermitian if for every vector field X on M we have

$$(\nabla_X(s_1), s_2) + (s_1, \nabla_X(s_2)) = X(s_1, s_2).$$

We are now equipped to define θ . Define a local trivialisation of L to be a section s_0 such that (s_0, s_0) is the identity function on M . Any section s can be written

$$s = fs_0$$

for some complex function f . Provided a vector field X we then define $\theta(X)$ to be the function such that

$$\nabla_X(s_0) = -i\theta(X)s_0$$

holds. Defined in this way, $\theta(X)$ is a one-form on M , and using the product rule for covariant derivatives we recover

$$\begin{aligned} \nabla_X(fs_0) &= X(f)s_0 + f\nabla_X(s_0) \\ &= (X(f) - i\theta(X)f)s_0. \end{aligned}$$

The one-form $\theta(X)$ is referred to as the connection one form associated to the local isometric trivialisation s_0 .

The curvature two-form may then be defined as:

$$\omega(X, Y)s = i(\nabla_X\nabla_Y - \nabla_Y\nabla_X - \nabla_{[X, Y]})(s). \quad (7.2)$$

Therefore locally, up to choice of local isometric trivialisation s_0 ,

$$\omega = d\theta$$

so ω is closed.

Lemma 7.1.2 remains true for non-trivial line bundles where Q_{pre} is defined in the same way as the cotangent bundle case and the space of smooth sections forms the representation space. This representation is usually reducible, in order to find an irreducible representation we introduce *polarisations*.

7.2 Polarisations

Definition 7.2.1. A polarisation P of a symplectic manifold M is a subbundle of $T^{\mathbb{C}}M$ such that

- P is integrable
- The fibre $P(m)$ is a maximal isotropic subspace of $T_m M$

where $T^{\mathbb{C}}\Omega$ is the complexified tangent bundle.

This definition requires some explanation, first note that the Frobenius integrability condition says

Theorem 7.2.1 (Frobenius integrability condition). *The following are equivalent:*

- $P \subset TM$ is integrable
- The vector space of vector fields $\hat{X}$ on M such that $\hat{X}(m) \in P(m)$ is a Lie subalgebra in the space of all vector fields on M .

Example 7.2.1. Consider the simple example of a vector field $\hat{X}$, where $P(x) = \hat{X}(x) \subset T_x M$. The existence of an integral curve through M where the vector tangent to the curve is $P(x)$ demonstrates the integrability of P .

Example 7.2.2. Consider the cotangent bundle $M = T^*Q$, where Q is an n -dimensional manifold. The cotangent bundle consists of pairs (p, q) where $q \in Q$ and p is an element of the cotangent space T_q^*Q . This is a manifold of dimension $2n$. Considering M as a vector bundle over Q , the fibres are then the individual cotangent spaces. The fibres foliate M , these are the surfaces with constant q .

Example 7.2.3. Continuing example 7.1.1, the set of all p directions is a polarisation called the vertical polarisation. The corresponding polarised space is:

$$\{g \in L^2(\mathbb{R}^2) \mid \nabla_{\partial/\partial p} g(p, q) = 0\}$$

as

$$\nabla_{\partial/\partial p} = \partial/\partial p - i\theta(\partial/\partial p) = \partial/\partial p - ipdq(\partial/\partial p) = \partial/\partial p$$

then

$$\{g \in L^2(\mathbb{R}^2) \mid \frac{\partial}{\partial p} g(p, q) = 0\}.$$

This space is simply the functions that are independent of the variable p , so the polarised space becomes $L^2(\mathbb{R})$.

Algebraic polarisations

There is a corresponding algebraic idea of polarisation which is often more practical to work with:

Definition 7.2.2. A complex algebraic polarisation of a functional $F \in \mathfrak{g}^*$ is a subalgebra $\mathfrak{h} \subset \mathfrak{g}_{\mathbb{C}}$ such that

- $F|_{[\mathfrak{h}, \mathfrak{h}]} = 0$ i.e. $X \mapsto 2\pi i \langle F, X \rangle$ is a character of $\mathfrak{h}$
- $\mathfrak{h}$ has maximal dimension, $\dim \mathfrak{h} = \frac{\dim \mathfrak{g} + \dim \text{stab}(F)}{2}$.

The maximal dimension criterion is the correlate of the maximal isotropic subspace requirement.

Theorem 7.2.2 (Kirillov, Kostant, Souriau, Auslander). *There is a bijection between the set of all G -invariant complex polarisations P of a coadjoint orbit $\Omega \subset \mathfrak{g}^*$ and the set of all admissible complex algebraic polarisations $\mathfrak{h}$ of a given element $F \in \Omega$.*

A real polarisation is then a subalgebra $\mathfrak{a} \subset \mathfrak{g}$ that satisfies the same properties. We can rephrase the Heisenberg group example seen in chapter 5 in the language of geometric quantisation using real polarisations.

Example 7.2.4. Let H be the Heisenberg group, $\mathfrak{a} = Y \oplus Z$ a subalgebra of $\mathfrak{h}$ subordinate to the functional F , ρ a one dimensional representation of $A = \exp \mathfrak{a}$, $\Omega \cong \mathbb{R}^2$ a 2-dimensional coadjoint orbit passing through F and $L = H \times_A \mathbb{C}$ the associated line bundle over $H/A \cong \mathbb{R}$. The pre-quantisation space is the space of sections $\Gamma(L) \cong \{f \in C^\infty(H) \mid f(xa) = \rho(a)^{-1} f(x)\}$. Let $\exp W = a \in A$, then the equivariance condition becomes

$$f(xa) = e^{2\pi i \langle F, X \rangle} f(x),$$

Letting $H \cong \mathbb{R}^3$ and $A \cong \mathbb{R}^2$, the equivariant functions can be written as $f(x, y, z)$, and $a = (0, y', z')$,

$$a \cdot f(x, y, z) = f(x, y + y', z + z' + xy') = e^{2\pi i \langle F, y'Y + z'Z \rangle} f(x, y, z) = e^{2\pi i \lambda z'} f(x, y, z).$$

Thus the equivariant functions vary in the y and z variables only up to the action of the subalgebra, and therefore can be specified by the variable x alone, the polarised space is then

$$\{f \in C^\infty(\mathbb{R})\}.$$

Taking the completion of this space with respect to the Lebesgue measure, we recover the representation space

$$f \in L^2(\mathbb{R}).$$

Holomorphic polarisations

A general Lie group admits a variety of polarisations, for compact groups Kähler polarisations are of particular interest:

Definition 7.2.3 (Kähler polarisation). Given the subbundle $P \subset T\Omega$ that uniquely corresponds to the complex algebraic polarisation $\mathfrak{p}$, we call P a Kähler polarisation when

- $\mathfrak{p} + \overline{\mathfrak{p}} = \mathfrak{g}_{\mathbb{C}}$
- $\mathfrak{p} \cap \overline{\mathfrak{p}} = \mathfrak{h}_{\mathbb{C}}$.
- $h(u, v) := \frac{1}{2i} \omega_F(u, \overline{v})$ is a positive definite Hermitian form on $\mathfrak{g}/\mathfrak{p}$.

where $\mathfrak{h} = \text{stab}(F)$.

A Kähler polarisation gives rise to a complex structure on the orbit Ω .

Almost complex structure

Definition 7.2.4. An almost complex structure on the tangent space of a manifold M is a mapping $J : TM \rightarrow TM$ that satisfies $J^2 = -1$, where 1 is the identity map.

We can then write the polarisation P and its conjugate as the eigenspaces:[25, §1.1]

$$P = \{v \in T^{\mathbb{C}}\Omega \mid Jv = -iv\},$$

and

$$\overline{P} = \{v \in T^{\mathbb{C}}\Omega \mid Jv = iv\},$$

conversely, we may define J via the polarisation. Any symplectic manifold of dimension $2n$ admits local coordinates $\{p_1, \dots, p_n, q_1, \dots, q_n\}$. Imbuing a symplectic manifold with an almost complex structure allows us to choose complex coordinates $z_j = p_j + iq_j$, $\overline{z}_j = p_j - iq_j$ [27, §5]. We can then write the subbundle $P = \text{span}\{\frac{\partial}{\partial \overline{z}_j}\}$ to denote the polarisation. The polarisation splits the complexified tangent space into an anti-holomorphic and holomorphic part

$$T_{\mathbb{C}}O_{\lambda} = P \oplus \overline{P}.$$

We call a section *holomorphic* if the derivative vanishes in the anti-holomorphic directions. The polarisation P corresponds to the anti-holomorphic directions.

The polarisation condition on the sections

$$\nabla_X s = 0 \quad X \in \mathfrak{p},$$

ensures that the sections are constant in the anti-holomorphic directions. Thus the polarised space becomes

$$\Gamma_{\text{hol}}(E_{\lambda}) = \{s \in \Gamma(E_{\lambda}) \mid s \text{ is holomorphic}\}.$$

We can make this space into a Hilbert space by imposing an L^2 condition and defining an inner product. Let the measure μ_Ω be the top degree form associated with the canonical symplectic form ω on the orbit so that $\mu = \omega^{\frac{n}{2}}$, then the L^2 condition is as expected:

$$f \in L^2(\Omega) \iff \left(\int_{\Omega} |f(\beta)|^2 d\mu \right)^{\frac{1}{2}}$$

Descriptions of the polarised space

We can describe the polarised space in two equivalent ways:

- The equivariant functions V_λ on the group (as in section 3.4), subject to $\hat{X} \cdot f - 2\pi i \langle F, X \rangle f = 0$ for X in the polarisation.
- The space of sections of the line bundle subject to the vanishing condition $\nabla_X s = 0$

Further, to an element X of the polarisation, we associate the fundamental vector field $\hat{X}$, evaluating this vector field on $f \in V_\lambda$:

$$\begin{aligned} \hat{X} \cdot f(y) &= \frac{d}{dt} f(y \exp tX) \Big|_{t=0} = \frac{d}{dt} e^{2\pi i \langle F, tX \rangle} f(y) \Big|_{t=0} \\ &= 2\pi i \langle F, X \rangle f(y). \end{aligned}$$

Therefore on the Lie algebra, the equivariance condition takes the form

$$(\hat{X} - 2\pi i \langle F, X \rangle) f(y) = 0.$$

Let $f \in V_\lambda$, and $X \in \text{stab}(\lambda)$. Using the equivariance of f :

$$\begin{aligned} \hat{X} \cdot f(g) &= \frac{d}{dt} f(\exp tXg) \Big|_{t=0} - 2\pi i \langle \lambda, X \rangle f(g) \\ &= \frac{d}{dt} e^{2\pi i \langle \lambda, tx \rangle} f(g) \Big|_{t=0} - 2\pi i \langle \lambda, X \rangle f(g) \\ &= 2\pi i \langle \lambda, x \rangle f(g) - 2\pi i \langle \lambda, X \rangle f(g) = 0 \end{aligned}$$

so that $\hat{X} \cdot f(g) = 0$ for $X \in \text{stab}(\lambda)$.

7.3 Hamiltonian G-manifolds

Hamiltonian G-manifolds

A Hamiltonian G-manifold is a symplectic manifold (M, ω) with a smooth action of a Lie group G that preserves the symplectic form. The action of the Lie group on the manifold defines a symmetry of the system, where some observable (i.e. a function $f \in C^\infty(M)$) is

constant under the action of the Lie group. The moment map identifies this function. Given a Lie algebra element X the vector field

$$\hat{X}(m) = \left. \frac{d}{dt} \right|_{t=0} \exp(tX) \cdot m$$

defines the infinitesimal action of the group. Let $\iota_{\hat{X}}\omega$ denote the contraction of ω by $\hat{X}$, the action of G is called Hamiltonian if ω is closed, in which case there exists a function (a zero form) H such that $\iota_{\hat{X}}\omega = dH$.

Definition 7.3.1 (Moment map). [35, Definition 6] The moment map for the symplectic action of G on M is a G -equivariant map

$$\mu : M \rightarrow \mathfrak{g}^*,$$

such that, for every $X \in \mathfrak{g}$, the vector field $\hat{X}$ is the Hamiltonian vector field of the function

$$m \mapsto \langle \mu(m), X \rangle.$$

The pair (ω, μ) satisfies the equation

$$d\langle \mu, X \rangle = \iota_{\hat{X}}\omega,$$

where $\langle \mu, X \rangle$ is understood as the function $\langle \mu, X \rangle(m) = \langle \mu(m), X \rangle$.

The function $\mu^X := \langle \mu(m), X \rangle$ then defines the observable conserved under the symmetries of the group action.

Example 7.3.1. Consider the situation as in example 7.1.1. Let $G = U(1)$ the group of unitary matrices, with Lie algebra $\mathfrak{g} \cong \mathfrak{g}^* = i\mathbb{R}$. Let $X \in \mathfrak{g}$, with vector field

$$\hat{X}((p, q)) = \left. \frac{d}{dt} \right|_{t=0} \exp(itX) \cdot (p, q) = \left. \frac{d}{dt} \right|_{t=0} \begin{pmatrix} \cos(tX) & -\sin(tX) \\ \sin(tX) & \cos(tX) \end{pmatrix} \cdot (p, q) = -X(q, -p)$$

then in local coordinates

$$\hat{X}((p, q)) = -X \left(q \frac{\partial}{\partial p} - p \frac{\partial}{\partial q} \right).$$

The contraction of ω :

$$\begin{aligned} \iota_{\hat{X}}\omega(\hat{Y}) &= \omega(\hat{X}, \hat{Y}) = dp \wedge dq(\hat{X}, \hat{Y}) \\ &= dp(\hat{X})dq(\hat{Y}) - dp(\hat{Y})dq(\hat{X}) \\ &= -X(qdq(\hat{Y}) + pdp(\hat{Y})) \\ &= -d \left(X \frac{q^2 + p^2}{2} \right) (\hat{Y}) \end{aligned}$$

Therefore the requirement $d\langle\mu, X\rangle = \iota_{\hat{X}}\omega$ reads

$$d\langle\mu, X\rangle = -d\left(X\frac{q^2 + p^2}{2}\right)$$

then

$$\langle\mu(m), X\rangle = -\left(X\frac{q^2 + p^2}{2}\right).$$

Therefore moment map is given by

$$\begin{aligned}\mu : \mathbb{R}^2 &\rightarrow \mathbb{R} \\ (p, q) &\mapsto -\frac{q^2 + p^2}{2}\end{aligned}$$

7.4 Coadjoint orbits

In the case of a symplectic manifold that is a coadjoint orbit the above definitions become almost trivial. Let the symplectic manifold Ω be a coadjoint orbit of a compact Lie group G . There is a natural G -action on Ω , making it a homogeneous space $\Omega \cong G/Stab(F)$. We define the KKS form as above

$$\sigma_{\Omega}(F)(K_*(X)F, K_*(Y)F) = \langle F, [X, Y] \rangle$$

where σ_{Ω} is G invariant.

The KKS form can be derived from the Maurer-Cartan form θ on the Lie group, this can be found in [3, Chapter 1, §2.1]. The Maurer-Cartan form acts as the connection one form for the Hermitian connection on the line bundle in theorem 7.4.2.

As there exists a smooth G action on Ω (the coadjoint action), then for $X \in \mathfrak{g}$ there is a corresponding vector field on Ω given by

$$\hat{X}(m) = \frac{d}{dt}(\exp tX) \cdot m \Big|_{t=0} = X \cdot m = K_*(X)m = [X, m]$$

where the last equality is only true for semi-simple Lie algebras.

Thus the map from $X \rightarrow \hat{X}$ is given by the differential of the coadjoint representation. The moment map

$$\mu : \Omega \rightarrow \mathfrak{g}^*$$

is simply the inclusion map

$$\mu : \Omega \hookrightarrow \mathfrak{g}^*.$$

then for $F \in \Omega$ and $X \in \mathfrak{g}$:

$$\mu^X(F) = \langle \mu(F), X \rangle = \langle F, X \rangle.$$

We may also define the co-moment map as [31, §1.1]

$$\mu^* : \mathfrak{g} \rightarrow C^\infty(\Omega)$$

where $\mu^*(X) = \mu^X$.

Theorem 7.4.1. *The co-moment map is a homomorphism of Lie algebras $\mathfrak{g}$ and $C^\infty(\Omega)$ with the Poisson bracket,*

$$\mu^*([A, B]) = \{\mu^*(A), \mu^*(B)\}.$$

Proof.

$$\mu^*([A, B]) = \langle F, [A, B] \rangle = \sigma_\Omega(F)(K_*(A)F, K_*(B)F)$$

for all $F \in \Omega$.

The Poisson bracket on $C^\infty(\Omega)$ implies

$$\{\mu^*(A), \mu^*(B)\} = \sigma_\Omega(F)(\hat{A}, \hat{B})$$

where $\hat{A}$ is the Hamiltonian vector field associated to A . Using the fact noted above that the map $A \rightarrow \hat{A}$ is given by the differential of the coadjoint orbit we write

$$\sigma_\Omega(F)(K_*(A)F, K_*(B)F)$$

so we are done. $\square$

Thus, the moment map respects the Poisson bracket. We can then compose the co-moment map with the quantisation map such that after polarisation we can find an irreducible representation of the Lie algebra $\mathfrak{g}$.

$$\mathfrak{g} \xrightarrow{\mu^*} C^\infty(\Omega) \xrightarrow{Q} \Gamma(L).$$

The existence of such a line bundle L with connection that satisfies the curvature requirement is equivalent to the integrality of σ_Ω that is:

Theorem 7.4.2 ([3, Chapter 1, §2.4, Proposition 4]). *If*

$$\int_C \sigma_\Omega \in \mathbb{Z} \quad \text{for every geometric 2-cycle } C \in \Omega,$$

then there exists a line bundle over M with a Hermitian connection ∇ such that

$$\text{curv}(\nabla) = i\sigma_\Omega.$$

In the language of geometric quantisation this theorem says that the map Q_{pre} satisfies the quantisation requirements and there exists a Hermitian line bundle with connection ∇ such that

$$Q_{pre} : C^\infty(M) \rightarrow \Gamma(L)$$

is a homomorphism of Lie algebras, (where $\Gamma(L)$ is the space of sections of the line bundle), if and only if the manifold M is integral.

Motivation

We conclude by offering some *post factum* motivation for the prequantisation procedure in the case of a coadjoint orbit.

Example 7.4.1. Let G be a compact Lie group with Lie algebra $\mathfrak{g}$, and Ω be an integral coadjoint orbit with symplectic form σ_Ω . A natural way of finding a representation of $\mathfrak{g}$ might be to take the vector field $\hat{X}$ associated to $X \in \mathfrak{g}$ and define the operator $\nabla_{\hat{X}}$ on the space of sections of the line bundle over Ω with connection ∇ . Define the map:

$$\pi(X) = \nabla_{\hat{X}}.$$

The curvature of the connection is defined as:

$$R(\hat{X}, \hat{Y})s \equiv [\nabla_{\hat{X}}, \nabla_{\hat{Y}}]s - \nabla_{[\hat{X}, \hat{Y}]}s.$$

In geometric quantisation, we specifically look for a line bundle with connection that has curvature proportional to the KKS form σ_Ω , such that:

$$R(\hat{X}, \hat{Y}) = -i\sigma_\Omega(\hat{X}, \hat{Y}).$$

As such, π does not define a Lie algebra homomorphism:

$$[\pi(X), \pi(Y)] = \nabla_{[\hat{X}, \hat{Y}]} - i\sigma_\Omega(\hat{X}, \hat{Y}).$$

We require the mapping to account for this $-i\sigma_\Omega(\hat{X}, \hat{Y})$ factor. Define the moment map

$$f : \mathfrak{g} \rightarrow C^\infty(\Omega)$$

$$X \mapsto f_X$$

$$\text{where } f_X(F) := \langle F, X \rangle$$

where f_X is a function on the orbit satisfying

$$df_X = -\sigma_\Omega(\hat{X}, \cdot).$$

Redefine π to be:

$$\pi(X) = \nabla_{\hat{X}} + if_X,$$

then the commutator becomes:

$$[\pi(X), \pi(Y)] = [\nabla_{\hat{X}} + if_X, \nabla_{\hat{Y}} + if_Y]$$

$$[\pi(X), \pi(Y)] = [\nabla_{\hat{X}}, \nabla_{\hat{Y}}] + i\hat{X}(f_Y) - i\hat{Y}(f_X).$$

Using our definition for df_X , we have $\hat{X}(f_Y) = \sigma_\Omega(\hat{X}, \hat{Y})$.

$$[\pi(X), \pi(Y)] = [\nabla_{\hat{X}}, \nabla_{\hat{Y}}] + i\sigma_\Omega(\hat{X}, \hat{Y}) + i\sigma_\Omega(\hat{X}, \hat{Y})$$

Combining this with the curvature $-i\sigma_\Omega(\hat{X}, \hat{Y})s + \nabla_{[\hat{X}, \hat{Y}]}s \equiv [\nabla_{\hat{X}}, \nabla_{\hat{Y}}]s$:

$$[\pi(X), \pi(Y)] = \nabla_{[\hat{X}, \hat{Y}]} - i\sigma_\Omega(\hat{X}, \hat{Y}) + 2i\sigma_\Omega(\hat{X}, \hat{Y})$$

$$[\pi(X), \pi(Y)] = \nabla_{[\hat{X}, \hat{Y}]} + i\sigma_\Omega(\hat{X}, \hat{Y})$$

Notice

$$f_{[X, Y]}(F) = \langle F, [X, Y] \rangle = \sigma_\Omega(F)(\hat{X}, \hat{Y})$$

then

$$[\pi(X), \pi(Y)] = \nabla_{[\hat{X}, \hat{Y}]} + if_{[X, Y]} = \pi([X, Y])$$

Therefore, π is a Lie algebra homomorphism.

Chapter 8

SU(2)

"Two stars keep not their motion in
one sphere"

Shakespeare, Henry VI

We take $SU(2)$ as our primary example of the orbit method applied to compact groups, much of this material can be found in any Lie theory textbook (see [18]) or orbit method treatment (see [3] or the introductory material in [22, pp. ~ 42-44]). $SU(2)$ is the group of 2×2 unitary matrices with determinant 1. Write $SU(2)$ as

$$SU(2) = \left\{ \begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix} \mid |\alpha|^2 + |\beta|^2 = 1, \alpha, \beta \in \mathbb{C} \right\}.$$

The Lie algebra consists of all traceless skew hermitian matrices,

$$\mathfrak{su}(2) = \left\{ \begin{pmatrix} ix_3 & x_1 + ix_2 \\ -x_1 + ix_2 & -ix_3 \end{pmatrix} \mid (x_1, x_2, x_3) \in \mathbb{R}^3 \right\}.$$

The matrices:

$$X_1 = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, X_2 = \frac{1}{2} \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, X_3 = \frac{1}{2} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

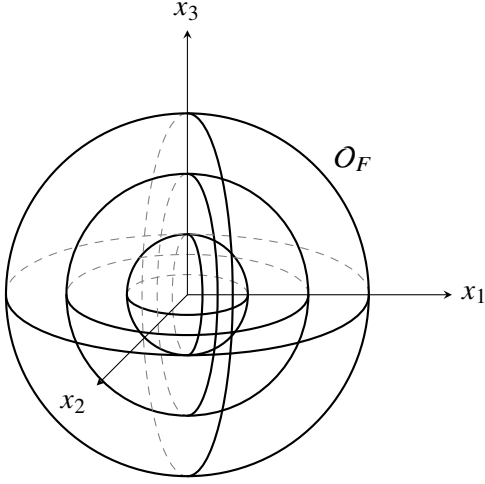
form a basis for $\mathfrak{su}(2)$.

We can make a vector space isomorphism between $\mathfrak{su}(2)$ and $\mathbb{R}^3$ via

$$(x_1 X_1, x_2 X_2, x_3 X_3) \mapsto (x_1, x_2, x_3).$$

8.1 Coadjoint orbits

For any semisimple Lie algebra we can associate the Lie algebra with the dual using the *Killing form*, a bilinear form invariant under the adjoint action. To $F \in \mathfrak{g}^*$ associate $X_F \in \mathfrak{g}$

Figure 8.1: (Co)adjoint orbits of $SU(2)$

via $\langle F_X, Y \rangle = (X, Y)_G$ [3, Chapter 5, §2.1]. Under this association, the coadjoint and adjoint orbits coincide, so the coadjoint action $K(g)$ is simply conjugation by g .

Recall the following theorem from linear algebra

Theorem 8.1.1. *Any normal matrix H is unitarily equivalent to a diagonal matrix. That is, there exists a unitary matrix U such that*

$$UHU^{-1} = D$$

As skew-Hermitian implies normality, we can deduce that any orbit O_F will pass through a diagonal element in $\mathfrak{su}(2)$,

$$x_3 \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}.$$

After our association of $\mathfrak{su}(2)$ with $\mathbb{R}^3$ and letting $SU(2)$ act on $\mathbb{R}^3$, we see that the orbits are spheres centered at the origin that pass through the point x_3 .

In the dual basis we can write the orbits as

$$\Omega_F = \{F_{a,b,c} \mid a^2 + b^2 + c^2 = r^2\}$$

defining spheres of radius r in $\mathbb{R}^3$.

Symplectic form

The symplectic form σ_Ω associated to an orbit Ω_F acts on the two fold tangent space $T\Omega_F \times T\Omega_F$. Define the map

$$\begin{aligned} \Phi_\xi : G &\rightarrow \mathfrak{g}^* \\ g &\mapsto K(g)\xi \end{aligned}$$

where $\Phi_\xi(e) = \xi$, Taking the differential of this map,

$$(\Phi_\xi)_*(X) = \left. \frac{d}{dt} \right|_0 \Phi_\xi(\exp(X \cdot t))$$

we see that $(\Phi_\xi)_*(X)$ is a vector tangent to ξ , that is $(\Phi_\xi)_*(X) \in T_\xi \Omega_\xi$. This vector can be thought to be "pointing in the X direction" and we write a basis ∂_{X_i} . In order to generate the basis vectors for the tangent space $T_\xi \Omega_\xi$ we take the differentials of the basis vectors in the Lie algebra.

The dual $\mathfrak{su}(2)^*$ matrices can be written:

$$F_{a,b,c} = \begin{pmatrix} -ic & -a - ib \\ a - ib & ic \end{pmatrix}$$

with basis

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, B = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}, C = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}.$$

The differential of the coadjoint representation is given by

$$K_*(X)F = [F, X].$$

To calculate the vector fields induced by the basis vectors take

$$\begin{aligned} K_*(X)F = [F, X] &= \frac{1}{2} \left[\begin{pmatrix} -(ic) & -a - ib \\ a - ib & ic \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} -(ic) & -a - ib \\ a - ib & ic \end{pmatrix} \right] \\ &= \begin{pmatrix} ib & -ic \\ -ic & -ib \end{pmatrix} \\ &= bC - cB \end{aligned}$$

Write the vector field

$$K_*(X) = b\partial_c - c\partial_b,$$

where ∂_a denotes a small change in the A direction. The other two vector fields are found in the same way and we omit the calculation:

$$K_*(X) = b\partial_c - c\partial_b, \quad K_*(Y) = c\partial_a - a\partial_c, \quad K_*(Z) = a\partial_b - b\partial_a$$

these vector fields are tangent to the orbit Ω_F .

We are now prepared to write the symplectic form:

$$\sigma_\Omega(F)(K_*(X)F, K_*(Y)F) = \langle F, [X, Y] \rangle$$

A two-form in local coordinates can be written as a C^∞ linear combination of the terms $\{da \wedge db, db \wedge dc, dc \wedge da\}$, therefore write:

$$\sigma_\Omega(F) = f_1 da \wedge db + f_2 db \wedge dc + f_3 dc \wedge da.$$

We can solve for the functions $f_i \in C^\infty$ by setting

$$\begin{aligned}
\langle F, [X, Y] \rangle &= (f_1 da \wedge db + f_2 db \wedge dc + f_3 dc \wedge da)(K_*(X)F, K_*(Y)F) \\
&= f_1 (da(K_*(X)F)db(K_*(Y)F) - da(K_*(Y)F)db(K_*(X)F)) \\
&\quad + f_2 (db(K_*(X)F)dc(K_*(Y)F) - db(K_*(Y)F)dc(K_*(X)F)) \\
&\quad + f_3 (dc(K_*(X)F)da(K_*(Y)F) - dc(K_*(Y)F)da(K_*(X)F)) \\
&= f_1 \left(da(b\partial_c - c\partial_b)db(c\partial_a - a\partial_c) - da(c\partial_a - a\partial_c)db(b\partial_c - c\partial_b) \right) \\
&\quad + f_2 \left(db(b\partial_c - c\partial_b)dc(c\partial_a - a\partial_c) - db(c\partial_a - a\partial_c)dc(b\partial_c - c\partial_b) \right) \\
&\quad + f_3 \left(dc(b\partial_c - c\partial_b)da(c\partial_a - a\partial_c) - dc(c\partial_a - a\partial_c)da(b\partial_c - c\partial_b) \right) /
\end{aligned}$$

Using $dj(\partial i) = \delta_j^i$ we find

$$\sigma_\Omega(F)(K_*(X)F, K_*(Y)F) = f_1 c^2 + f_2 c \cdot a + f_3 b \cdot c.$$

An easy calculation then shows

$$\langle F, [X, Y] \rangle = \langle F, Z \rangle = c$$

So,

$$c = f_1 c^2 + f_2 c \cdot a + f_3 b \cdot c$$

Proceeding similarly for the other combinations of basis vector inputs $K_*(X)F, K_*(Z)F$ and $K_*(Y)F, K_*(Z)F$, we reach the following set of equations

$$\begin{aligned}
f_1 c \cdot a + f_2 a^2 + f_3 a \cdot b &= a \\
f_1 c \cdot b + f_2 b \cdot a + f_3 b^2 &= b \\
f_1 c^2 + f_2 c \cdot a + f_3 b \cdot c &= c.
\end{aligned}$$

This reduces to the requirement

$$f_1 c + f_2 a + f_3 b = 1 \quad \forall (a, b, c) \in \Omega_F.$$

Recall that this form is defined per orbit, that is spheres in $\mathbb{R}^3$, so we may take

$$f_1 = c, \quad f_2 = a, \quad f_3 = b$$

to see

$$c^2 + a^2 + b^2 = 1.$$

This condition must be true for an orbit of any radius, so divide by the radius squared to find

$$f_1 = \frac{c}{a^2 + b^2 + c^2}, \quad f_2 = \frac{a}{a^2 + b^2 + c^2}, \quad f_3 = \frac{b}{a^2 + b^2 + c^2}.$$

So write the canonical symplectic form as:

$$\sigma_\Omega(F) = \frac{cda \wedge db + adb \wedge dc + bdc \wedge da}{a^2 + b^2 + c^2}. \quad (8.1)$$

This is proportional to the the standard volume form on the 2-sphere.

8.2 Integral orbits

Now equipped with the canonical form on the orbits, we are prepared to understand the the second equivalent statement of the integrality condition 6.3.1 as it relates to $SU(2)$:

$$\int_C \sigma \in \mathbb{Z}$$

for every singular 2-cycle C in Ω .

2-cycles in an orbit Ω_r

To calculate the singular 2-cycles in an orbit $\Omega_r \cong S^2$, we use the fact

$$H_2(S^2) \cong \mathbb{Z}.$$

This means the second homology group is one-dimensional and hence generated by one element. We associate this "basis-chain" with the map that covers the whole sphere and conclude that the 2-cycles in Ω_r are given by the sphere (the whole orbit) itself. The full extent of the justification of this fact is beyond the current scope, see section 9.2 for further detail.

With this in mind we can write the integral

$$\int_{\Omega_r} \frac{cda \wedge db + adb \wedge dc + bdc \wedge da}{a^2 + b^2 + c^2}.$$

This is the integral of the constant function in $\mathbb{R}^3$ over the sphere, moving to spherical coordinates we find the symplectic volume is given by

$$\text{vol}(\Omega_r) = 4\pi r,$$

thus integral is integer valued for all orbits of radius $r = \frac{1}{4\pi}\mathbb{Z}$. Therefore the set of integral orbits is given by

$$\mathcal{O}_{int} = \{\Omega_r \in \mathcal{O}(K) \mid r \in \frac{1}{4\pi}\mathbb{Z}\}.$$

8.3 Characters

The Kirillov character formula for compact groups takes the form

$$\mathrm{tr} \pi_\lambda \exp(X) = \frac{1}{j(X)} \int_{\Omega_\lambda} e^{2\pi i \langle F, X \rangle} \sigma$$

In $SU(2)$ this is the Fourier transform of the surface measure on the sphere of radius r .

It can be shown using the Weyl character formula (see [3, Chapter 5, Theorem 1]) that the character of the irreducible representation with highest weight r is given by

$$\chi_r(\exp zZ) = \frac{\sin((r+1)z/2)}{\sin(z/2)} \quad (8.2)$$

The characters for $SU(2)$ are uniquely determined by their values on the maximal torus, these are the diagonal elements $\exp zZ$. Using the Kirillov character formula we can calculate

$$\int_{\Omega_{rw}} e^{2\pi i \langle F, X \rangle} \sigma = \frac{r}{4\pi} \int_0^{2\pi} d\phi \int_{-\pi/2}^{\pi/2} e^{\frac{1}{2}rz \sin \theta} \cos \theta d\theta = \frac{2}{z} \sin \frac{rz}{2}$$

The square root of the Jacobian of the exponential map equates to:

$$j(zZ) = \frac{\sin(z/2)}{z/2}.$$

We conclude:

$$\chi_r(\exp zZ) = \frac{2}{z} \sin \frac{rz}{2} \cdot \left(\frac{\sin(z/2)}{z/2} \right)^{-1} = \frac{\sin(rz/2)}{\sin(z/2)}. \quad (8.3)$$

Comparing equations 8.2 and 8.3, the character produced by the Kirillov formula is one weight displaced. To rectify this correspondence we add a factor of 1. For a general compact group this correction is called the *metaplectic correction*, see [3, Chapter 5, §3.1] for more detail. For $SU(2)$ the character formula then reads

$$\mathrm{tr} \pi_\lambda \exp(X) = \frac{1}{j(X)} \int_{\Omega_{\lambda+\rho}} e^{2\pi i \langle F, X \rangle} \sigma. \quad (8.4)$$

8.4 Construction of representations

This section details the method outlined in § 6.6 applied to $SU(2)$, using parts of the derivation in [15]. First consider the naïve application of the orbit method procedure as in the Heisenberg group example.

Lemma 8.4.1. *Given a compact Lie algebra $\mathfrak{g}$ and its dual $\mathfrak{g}^*$, then every element of $\mathfrak{g}$ is conjugate to the maximal torus, as such the stabiliser of a regular element $\lambda \in \mathfrak{g}^*$ is the maximal torus T :*

$$\begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}.$$

The only non regular element of $SU(2)$ corresponds to the 0 orbit which corresponds to the trivial representation, thus we restrict our attention to regular orbits for the remainder of this section, appending the trivial representation to any results. The torus T is one dimensional and parameterised by $\theta \in \mathbb{R}$. As in section 8.3 the characters of T are given by

$$\begin{aligned}\rho_\lambda : T &\rightarrow \mathbb{C} \\ \theta &\mapsto e^{in\theta}\end{aligned}$$

where λ satisfies $e^{2\pi i \langle \lambda, X \rangle} = e^{in\theta}$ for $\exp X \in T$ and X is an element of the Cartan subalgebra parameterised by θ . We then form the space

$$V = \{f : SU(2) \rightarrow \mathbb{C} \mid f(xh) = \rho(h)^{-1}f(x)\}$$

for $h \in T$ and $x \in SU(2)$, where G acts on the left $g \cdot f(x) = f(g^{-1} \cdot x)$ to form a representation of the group (or equivalently a G -module). We will assume the group action on this module is determined by the character $\rho(h) = e^{-in\theta}$. Let L_n denote the associated line bundle $SU(2) \times \mathbb{C} / \sim$ over $SU(2)/T$. The space V is then equivalent to the space of sections of L_n denoted $\Gamma(L_n)$ see 3.4. The orbit method then requires that we consider the principal bundle $\pi : SU(2) \rightarrow SU(2)/T$ with sections $s : SU(2)/T \rightarrow SU(2)$. As manifolds there exist diffeomorphisms $SU(2) \cong S^3$ and $SU(2)/T \cong S^2$, so the bundle may be written

$$\pi : S^3 \rightarrow S^2,$$

this is called the Hopf bundle or Hopf fibration of the 3-sphere over the 2-sphere. We then solve the master equation

$$s(x)g = h(x, g)s(x \cdot g),$$

then the induced representation will be of the form

$$(\text{Ind}_H^G \rho)(g)(f)(x) = A(x, g)f(x \cdot g)$$

where $A(x, g) = \rho(h(x, g))$.

This induced representation acting on V is not necessarily irreducible, however as we will see, restriction to the holomorphic sections $\Gamma_{hol}(L_n)$ is exactly the right requirement to recover the irreducible representations. The holomorphic condition is why this method of induction is sometimes referred to as holomorphic induction. As seen in the chapter on geometric quantisation, the holomorphic condition is a consequence of the polarisation.

Polarisation

Recall that the polarisation $\mathfrak{p}$ is a subalgebra of the complexified Lie algebra. The complexification of $\mathfrak{su}(2)$ is $\mathfrak{sl}(2, \mathbb{C})$ which decomposes as:

$$\mathfrak{h} : \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \mathfrak{n}_+ : \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \mathfrak{n}_- : \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

For a compact Lie group, the standard choice for a polarisation $\mathfrak{p}$ is the Borel subalgebra corresponding to a choice of positive roots Δ^+ :

$$\mathfrak{p} = \mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}_+$$

then

$$\mathfrak{p} = \begin{pmatrix} a & b \\ 0 & -a \end{pmatrix} \quad \forall a, b \in \mathbb{C}.$$

Following the exposition of the general theory (see §7.2), the polarised space is written:

$$V_{pol} = \{f \in V_\lambda \mid (\hat{X} - 2\pi i \langle F, X \rangle)f = 0 \quad \forall P \in \mathfrak{p}\}$$

Due to the equivariance condition, when $H \in \text{stab}(\lambda)$ then $H \cdot f = 0$, as any $P \in \mathfrak{p}$ can be written as $H + X$ where $H \in \mathfrak{h}$ and $X \in \mathfrak{n}_+$, the polarised space becomes:

$$\{f \in V_\lambda \mid X \cdot f = 0 \quad \forall X \in \mathfrak{n}_+\}.$$

The action of X on f is given by the fundamental vector field associated to X :

$$(\hat{X} \cdot f)(g) = \left. \frac{d}{dt} \right|_{t=0} f(g \cdot \exp(tX))$$

this action extends $\mathbb{C}$ linearly to $\mathfrak{sl}(2, \mathbb{C})$. As seen in chapter 7, the polarised space may then equivalently be written as

$$\Gamma_{pol}(L_n) = \left\{ s \in \Gamma(L_n) \mid \nabla_X s = 0 \quad \forall X = \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix} \right\}$$

Complex structure and holomorphicity

The polarisation defines a complex structure on the orbit, where the complexified tangent space at the origin splits as:

$$T^{\mathbb{C}}\Omega_\lambda = \mathfrak{sl}(2, \mathbb{C})/\text{stab}(\lambda)^{\mathbb{C}} = \mathfrak{n}_- \oplus \mathfrak{n}_+$$

The polarisation dictates that the antiholomorphic directions are given by $\mathfrak{n}_+$. The polarisation requirement $X \cdot f = 0$ for $X \in \mathfrak{n}_+$ means that the derivative of f must vanish in the anti-holomorphic directions on the orbit. The function f uniquely determines a section s of the line bundle over the orbit.

A local section is a map

$$\sigma : U_0 \rightarrow SU(2)$$

where U_0 is an open set in $SU(2)/T$ and $\pi(\sigma(x)) = x$.

The function $X \cdot f$ is again an equivariant function in V_λ so also descends to a section s of the associated line bundle. The section s has local trivialisation $X \cdot f(z) = (X \cdot f)\sigma_0(z)$ on the chart $U_0 \cong \mathbb{C}$ where $\sigma_0 : U_0 \rightarrow SU(2)$. The condition $X \cdot f = 0$ then shows

$$X \cdot f(z) = (X \cdot f)\sigma_0(z) = 0$$

So local trivialisation of the section s corresponding to f , given by the pull back over the local trivialisation σ will have 0 derivative in the antiholomorphic direction in the tangent space. Given a section $s \in \Gamma(L_n)$ where $\nabla_X s = 0$ for $X \in \mathfrak{n}_+$, then by definition of the Cauchy-Riemann operator

$$\frac{\partial}{\partial \bar{z}} \tilde{f} = 0$$

for the local trivialisation $\tilde{f}$ of s . The vector field generated by the action of $X \in \mathfrak{n}_+$ on the orbit $SU(2)/T$ precisely coincides with the antiholomorphic directions in the tangent space, hence its vanishing is equivalent to the Cauchy-Riemann operator.

We then call the section s a holomorphic section if the local trivialisations are holomorphic on their respective charts. The representation is then given by

$$\Gamma_{hol}(L_n) = \{s \in \Gamma(L_n) \mid s \text{ is holomorphic}\} \quad (8.5)$$

8.5 Explicit calculation of representations

We now explicitly calculate the irreducible representations of $SU(2)$ recovering the homogeneous polynomials of degree n seen in the general theory. Let $f \in V_{pol}$ then as above we have for all $X \in \mathfrak{n}_+$,

$$X \cdot f = 0$$

and

$$f \left(\begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix} \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} \right) = e^{in\theta} f \left(\begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix} \right)$$

where the subgroup T acts via the character associated to the integer n . For

$$X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

the evaluation of the corresponding vector field at f is written

$$(\hat{X} \cdot f) \left(\begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix} \right) = \frac{d}{dt} \Big|_{t=0} f \left(\begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix} \cdot \exp \begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix} \right)$$

The matrix X does not exponentiate to an element of $SU(2)$, so $X \cdot f$ for $f \in C^\infty(SU(2))$ is not defined. As $X \in \mathfrak{sl}(2, \mathbb{C})$ we can define X in terms of the complexification of $\mathfrak{su}(2)$:

$$X = X_1 - iX_2.$$

We then define the action of the vector field associated to X as

$$\hat{X}f(g) = \frac{d}{dt} \Big|_{t=0} f(g \exp tX_1) - i \frac{d}{dt} \Big|_{t=0} f(g \exp tX_2).$$

By the chain rule, this corresponds to the directional derivative of f along the complexified tangent vector gX . We can evaluate this tangent vector directly by matrix multiplication:

$$\begin{aligned}
 gX &= g(X_1 - iX_2) \\
 &= \begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix} \frac{1}{2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} - i \begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix} \frac{1}{2} \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \\
 &= \frac{1}{2} \begin{pmatrix} -\beta & \alpha \\ -\bar{\alpha} & -\bar{\beta} \end{pmatrix} - \frac{i}{2} \begin{pmatrix} i\beta & i\alpha \\ i\bar{\alpha} & -i\bar{\beta} \end{pmatrix} \\
 &= \frac{1}{2} \begin{pmatrix} -\beta & \alpha \\ -\bar{\alpha} & -\bar{\beta} \end{pmatrix} + \frac{1}{2} \begin{pmatrix} \beta & \alpha \\ \bar{\alpha} & -\bar{\beta} \end{pmatrix} \\
 &= \begin{pmatrix} 0 & \alpha \\ 0 & -\bar{\beta} \end{pmatrix}.
 \end{aligned}$$

Therefore, the differential operator acting on f evaluates to:

$$(\hat{X}f)(g) = df_g(gX) = df_g \begin{pmatrix} 0 & \alpha \\ 0 & -\bar{\beta} \end{pmatrix}.$$

This means the vector field acts only on the second column variables, so only the variables in the second column contribute to the value of $X \cdot f$.

The holomorphism requirement $X \cdot f = 0$ then implies that the values in the second column must be fixed to ensure $X \cdot f = 0$. The function f then depends only on the first column of the matrix, $f(g \exp X) = f(g) = f\left(\begin{pmatrix} \alpha \\ -\bar{\beta} \end{pmatrix}\right)$.

Then given an element of the maximal torus:

$$\begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix} \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} = \begin{pmatrix} \alpha e^{i\theta} & \beta e^{-i\theta} \\ -\bar{\beta} e^{i\theta} & \bar{\alpha} e^{-i\theta} \end{pmatrix}$$

the equivariance condition says:

$$f\left(\begin{pmatrix} \alpha e^{i\theta} & \beta e^{-i\theta} \\ -\bar{\beta} e^{i\theta} & \bar{\alpha} e^{-i\theta} \end{pmatrix}\right) = e^{in\theta} f\left(\begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix}\right).$$

Using these two facts together, write f as a function of two variables that satisfies the equivariance condition:

$$f\left(\begin{pmatrix} \alpha e^{i\theta} \\ -\bar{\beta} e^{i\theta} \end{pmatrix}\right) = e^{in\theta} f\left(\begin{pmatrix} \alpha \\ -\bar{\beta} \end{pmatrix}\right).$$

We require functions $f(x_1, x_2)$ of two variables satisfying the equivariance and holomorphicity conditions. Let $\alpha = x_1$ and $-\bar{\beta} = x_2$ and make the ansatz that the $f(x_1, x_2)$ are homogeneous polynomials:

$$P(x_1, x_2) = \sum_{j=0}^n a_j x_1^j x_2^{n-j}.$$

These polynomials satisfy the equivariance condition:

$$P(e^{i\theta}x_1, e^{i\theta}x_2) = \sum_{j=0}^n a_j (e^{i\theta}x_1)^j (e^{i\theta}x_2)^{n-j} = e^{in\theta} P(x_1, x_2).$$

We know from the above that any function dependent on only the variables in the first column satisfies the holomorphicity condition so we conclude that the homogeneous polynomials satisfy both criteria.

The domain of these polynomials can be extended to $\mathbb{C}^2$ as the required holomorphicity and equivariance properties are independent of the relationship between the variables. Therefore we can conclude that the space V_{pol} of polarised ρ equivariant functions contains the homogeneous polynomials of degree n in two complex variables. In fact, these functions completely define the space V_{pol} .

We can take the analytic continuation of $f \in V_{pol}$ extending the domain from (x_1, x_2) (as defined above) to $(z_0, z_1) \in \mathbb{C}^2$ by defining h such that

$$h(wz_0, wz_1) = w^n h(z_0, z_1) \quad w \in \mathbb{C}$$

and

$$X \cdot h = 0$$

This function is analytic and $h = f$ over (x_1, x_2) .

$$h(z_0, z_1) = \sum_{i,j=0} c_{i,j} z_0^i z_1^j$$

The equivariance condition then implies:

$$\sum_{i,j=0} c_{i,j} w^{i+j} (z_0)^i (z_1)^j = w^n \sum_{i,j=0} c_{i,j} z_0^i z_1^j.$$

Equating coefficients observe that $i + j = n$ or $c_{i,j} = 0$, therefore

$$h = \sum_{i=0} c_i z_0^i z_1^{n-i}$$

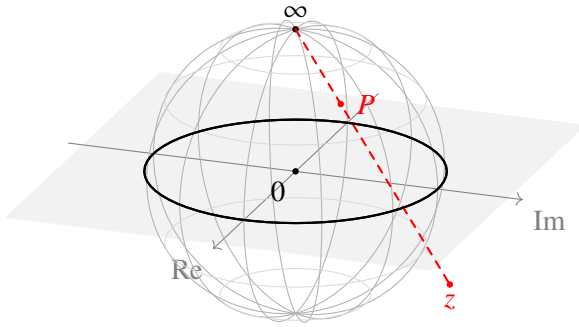
We then conclude that the space of holomorphic sections of the line bundle L_n and hence the space of polarised functions V_{pol} is equal to the space of homogeneous polynomials of degree n in two complex variables. The action of $SU(2)$ on this space is given by

$$(\pi_n(g)P)(z_0, z_1) = P(g^{-1}(z_0, z_1)) = P(\bar{\alpha}z_0 - \beta z_1, \bar{\beta}z_0 + \alpha z_1).$$

Borel-Weil

We conclude with a brief remark on the relationship to the Borel-Weil theory.

Every coadjoint orbit can be understood as a flag manifold $\mathcal{F} = G/B$ with complex structure, specifically a Kähler manifold. Given a line bundle L_λ over $\mathcal{F}$, with total space $G \times_B \mathbb{C}_\chi$, where B acts on $\mathbb{C}_\chi$ via the character χ , the Borel-Weil theorem states:

Figure 8.2: Riemann sphere $\mathbb{CP}^1$

Theorem 8.5.1. Borel - Weil From [3, Chapter 5, §3.4, Theorem 7]

The space $\Gamma_{Hol}(L_\lambda)$ is non-zero exactly when λ is a dominant, integral weight. In this case G acts on $\Gamma_{Hol}(L_\lambda)$ via an irreducible representation π_λ with highest weight λ .

The line bundle L_λ is guaranteed to exist by the integrality condition and this representation is exactly the space of square integrable holomorphic sections of the Riemann sphere (fig. 8.2), Γ_λ in equation 8.5.

Chapter 9

Appendix

Covering Spaces

Definition 9.0.1. Given a space E the map $f : E \rightarrow B$ is called a covering of a topological space B by E if for every neighbourhood of a point $b \in B$, there exists $f^{-1}(U_b) \in E$ such that $f^{-1}(U_b)$ is homeomorphic to a disjoint union of copies of U_b .

Definition 9.0.2. A cover is called universal if the covering space E is simply connected

Theorem 9.0.1. *Inverse function theorem [33, Theorem 4.5] If given a differentiable map between differentiable manifolds $f : M \rightarrow N$, the differential $f_{*,p} : T_p M \rightarrow T_p N$ is a linear isomorphism at a point p in M then there exists an open neighbourhood U of p such that $f|_U$ is a diffeomorphism.*

Example 9.0.1. A simple example is given by $\mathbb{R}$ covering S^1 . Consider the map

$$\begin{aligned} f : \mathbb{R} &\rightarrow S^1 \\ t &\mapsto e^{2\pi i t} \end{aligned}$$

For a neighbourhood U_x of S^1 , $f^{-1}(U_x)$ will be the union of open sets in $\mathbb{R}$ separated by integer multiples of 2π . It is easy to see how this set is homeomorphic to a disjoint union of copies of U_x .

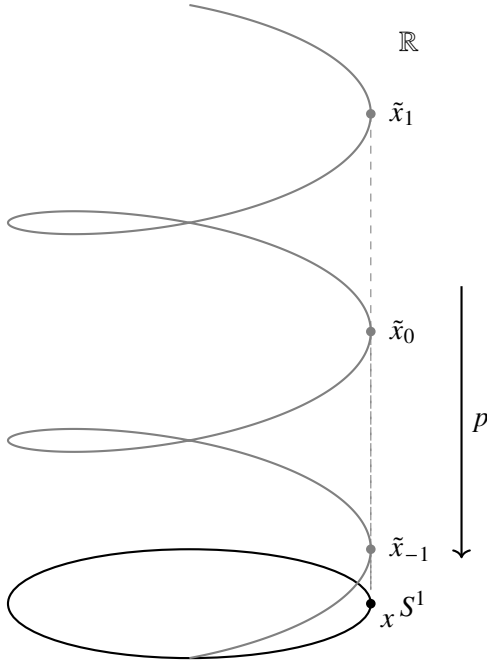
In the context of Lie groups a simple example is given by $SU(2)$ as the double cover of $SO(3)$

Example 9.0.2. $SU(2)$ is the double cover of $SO(3)$

Proof. The Lie algebra of $SU(2)$ can be associated with $\mathbb{R}^3$ ¹. The adjoint action of $SU(2)$ on $\mathfrak{su}(2)$ is an automorphism, then

$$\text{Ad}(g)X = gXg^{-1}.$$

¹Details in chapter 8.

Figure 9.1: $\mathbb{R}$ covering S^1

By definition the matrices in $SU(2)$ are unitary so Ad preserves the norm of X . Therefore the action of g under the adjoint representation corresponds to a rotation of $\mathfrak{su}(2) \cong \mathbb{R}^3$, therefore the image of Ad is contained in $SO(3)$. The adjoint representation of $SU(2)$ is irreducible, by Schur's lemma $\text{Ad}(g)X = X$ implies $g = \pm I$. Therefore $\text{Im}(\text{Ad}) \cong SU(2)/\{\pm I\}$. The Lie algebras $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ are isomorphic, therefore ad is an isomorphism in a neighbourhood of the identity and thus by translation an isomorphism over all of $\mathfrak{su}(2)$. By theorem 9.0.1 Ad is then a diffeomorphism and the image is open in $SO(3)$. The group $SU(2)$ is compact, as Ad is a continuous mapping the image must preserve compactness, a compact subset of $SO(3)$ is closed thus the image of Ad is closed. Therefore the mapping Ad is both open and closed therefore the map must be surjective. $\square$

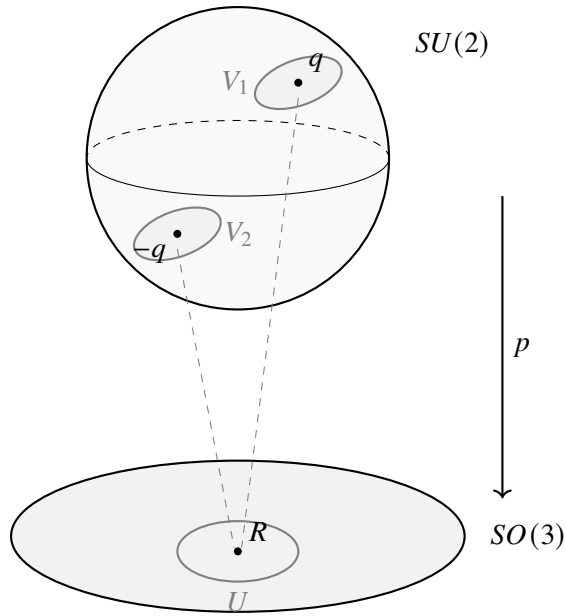
9.1 Semisimple Lie algebras

Following Cartan and Killing, the structure of semisimple Lie algebras is well understood. The idea is that any semisimple Lie algebra $\mathfrak{g}$ decomposes into a direct sum of simple Lie groups

$$\mathfrak{g} = \mathfrak{s}_1 \oplus \dots \oplus \mathfrak{s}_n.$$

Any simple Lie algebra over the complex numbers $\mathbb{C}$ is uniquely identified by its root system², which we understand through Dynkin diagrams. Through this classification of all

²See following chapter

Figure 9.2: $SU(2)$ covering $SO(3)$

simple Lie algebras over $\mathbb{C}$ we in turn gain an understanding of the semisimple groups. These definitions can be found in any Lie algebra textbook see [18] or chapter 3 of [9].

Definition 9.1.1. Simple Lie algebras
The Four Classical Infinite Families

Type	Complex Lie Algebra	Compact Real Group
$A_n (n \geq 1)$	$\mathfrak{sl}(n+1, \mathbb{C})$	$SU(n+1)$
$B_n (n \geq 2)$	$\mathfrak{so}(2n+1, \mathbb{C})$	$Spin(2n+1)$
$C_n (n \geq 3)$	$\mathfrak{sp}(2n, \mathbb{C})$	$Sp(n)$
$D_n (n \geq 4)$	$\mathfrak{so}(2n, \mathbb{C})$	$Spin(2n)$

The Five Exceptional Families

Type	Complex Lie Algebra	Compact Real Group
G_2	$\mathfrak{g}_2$	G_2
F_4	$\mathfrak{f}_4$	F_4
E_6	$\mathfrak{e}_6$	E_6
E_7	$\mathfrak{e}_7$	E_7
E_8	$\mathfrak{e}_8$	E_8

Compact simply connected case

In the study of Lie groups the simply connected compact groups are of particular utility and interest. Due to Elie Cartan, we may associate to any complex semisimple Lie algebra, a 'real form'. Using this construction, we can develop a correspondence between complex semisimple Lie algebras and simply connected compact Lie groups. Given a complex semisimple Lie algebra, Cartan showed this gives rise to a compact real form (i.e a compact Lie algebra such that its complexification is equal to the original semisimple Lie algebra). This compact real form in turn exponentiates to a simply connected compact Lie group. Now given a simply connected compact Lie group, its Lie algebra is always semisimple and compact, so we may simply take the complexification to find a corresponding semisimple Lie algebra.

Lemma 9.1.1. *Given a simply connected compact Lie group, its Lie algebra is always semisimple and compact*

Lemma 9.1.2. *Given a simply connected compact Lie group, its centre is finite*

If K is simply connected compact Lie group, then $\mathfrak{k}^{\mathbb{C}}$ is semisimple and every semisimple Lie algebra over $\mathbb{C}$ has a compact real form isomorphic to the Lie algebra of a simply connected compact Lie group.

9.2 Singular homology

Here we collect some definitions and intuitions regarding the homology theory of singular complexes. This material along with further detail for the interested reader can be found in [23] and a great set of lecture notes by Gereon Quick.

Definition 9.2.1. The standard n -simplex denoted by Δ^n is defined as the set

$$\Delta^n = \{(t_0, \dots, t_n) \in \mathbb{R}^n \mid \sum t_i = 1, \quad t_i \geq 0\}$$

Definition 9.2.2. A singular n -simplex is a continuous map

$$\sigma : \Delta^n \rightarrow X$$

where X is a topological space.

Example 9.2.1. The simplex Δ^1 can be understood as the set of points along the line from $(1, 0)$ to $(0, 1) \in \mathbb{R}^2$.

Take $X = \mathbb{R}^2$ and the singular 1-simplex $\sigma : \Delta^1 \rightarrow X$. Let the image of $\sigma = S^1 \subset \mathbb{R}^2$, with $\sigma((0, 1)) = \sigma((1, 0))$.

Definition 9.2.3. Define the set $Sing_n(X)$ to be the set of all n -simplices on X .

We then define the map

$$d_i^n : \text{Sing}_n(X) \rightarrow \text{Sing}_{n-1}(X)$$

Notice this map takes identifies end-points of a singular n -simplex with an $(n - 1)$ -simplex. We now introduce extra structure on $\text{Sing}_n(X)$ by letting S_n be the free abelian group generated by the singular n -simplices.

Definition 9.2.4. The group of singular n -chains $S_n(X)$ is defined to be the free abelian group generated by the singular n -simplices. An n -chain is an element

$$\sum_{i=1}^k \alpha_i \sigma_i$$

where $\alpha_i \in \mathbb{Z}$ and $\sigma_i \in \text{Sing}_n(X)$.

We can then define the boundary operator

$$\begin{aligned} \partial : S_n(X) &\rightarrow S_{n-1}(X) \\ \sum_{i=1}^k \alpha_i \sigma_i &\mapsto \sum_{i=1}^k \alpha_i \partial(\sigma_i) \end{aligned}$$

Where

$$\partial(\sigma) = \sum_{i=0}^n (-1)^i \sigma^{(i)}$$

Any n -simplex σ can be written as a list of ordered vertices:

$$\sigma = [v_0, v_1, \dots, v_n]$$

as such the boundary operator can be written:

$$\partial_n([v_0, v_1, \dots, v_n]) = \sum_{j=0}^n (-1)^j [v_0, \dots, \hat{v}_j, \dots, v_n].$$

where $\hat{v}_j$ refers to the removal of this term, resulting in an $n - 1$ simplex.

Finally we can define n -cycles,

Definition 9.2.5. An n -cycle is an n -chain c with $\partial c = 0$.

The group of all n -cycles is written as

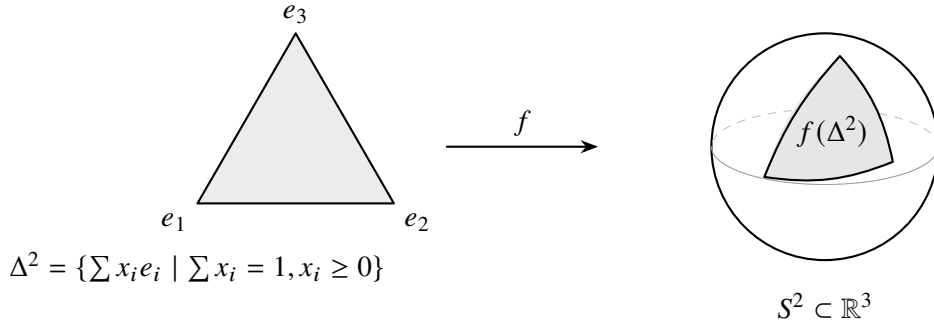
$$Z_n(X) := \{c \in S_n(X) \mid \partial(c) = 0\}$$

In short, cycles are chains with no boundary, the example considered above is a cycle.

The groups $S_n(X)$ of singular n -chains form a chain complex with boundary operator δ_n :

$$\dots \longrightarrow S_{n+1} \xrightarrow{\delta_{n+1}} S_n \xrightarrow{\delta_n} S_{n-1} \longrightarrow \dots \longrightarrow S_1 \xrightarrow{\delta_1} S_0 \xrightarrow{\delta_0} 0$$

The exactness of this sequence means $\partial^2 = 0$.

Figure 9.3: Mapping from standard 2-simplex Δ^2 to the sphere S^2

Calculating cycles, boundaries and homology groups

In order to calculate with these objects explicitly it is useful to use different homological theories, namely cellular homology or simplicial homology. For suitable topological spaces the isomorphism holds

$$H_n^\Delta(X) \cong H_n(X) \cong H_n^{CW}(X),$$

the interested reader is directed to the Eilenberg-Steenrod Axioms for more detail.

Integration of a differential form over a chain

As seen above, a k -chain is a formal sum of maps with coefficients in the integers from the standard k -simplex Δ^k to the manifold of interest X . Figure 9.3 illustrates that each chain maps the simplex onto a subset of the manifold.

Given a chain $c = \sum \sigma_i * \alpha_i$ where the σ_i are maps from the standard k -simplex Δ^k to the manifold X and a differential k -form ω , we define the integral of $\int_C \omega$ to be $\sum \alpha_i \int_{\Delta^k} (\sigma_i^* \omega)$. The term $\sigma_i^* \omega$ is the pullback of the differential form on X to a differential form on Δ^k the interior of which is an open subset of $\mathbb{R}^k$ and can hence be integrated in local coordinates. We define the pullback as follows, given a point $p \in \Delta^k$ and $\mathbf{v} \in T_p \Delta^k$,

$$(\sigma_i^* \omega)_p(\mathbf{v}) := \omega_{\sigma_i(p)}(\sigma_{i*}(\mathbf{v})).$$

The pullback allows us to define a differential form on the Euclidean plane that uniquely corresponds to the differential form ω on the manifold.

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